

Sri Lanka Institute of Information Technology



Research Project - IT4010

Check List 1-Git repository and Design Report

Group ID: 25-26J-473

Name	Registration Number
Luvinsion I	IT22323316
Krushanth A	IT22920218
Sangeerthan S	IT22346568
Fernandopulle S N	IT22000576

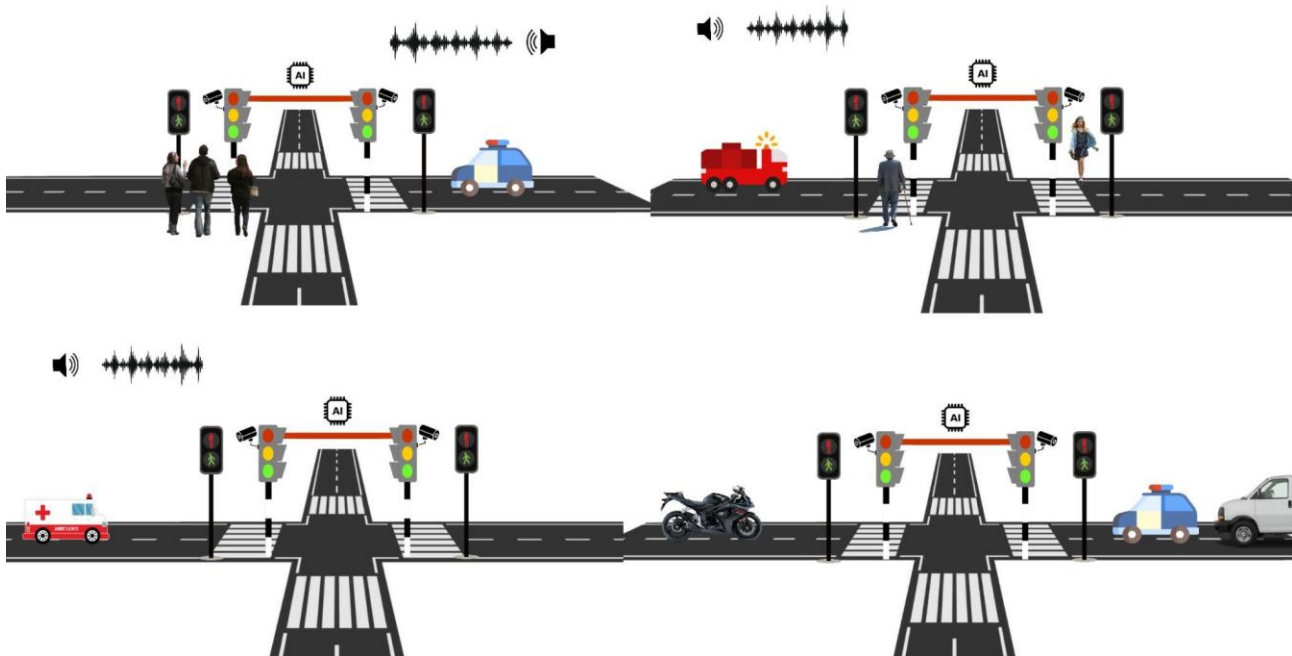
Table of Contents

Group ID: 25-26J-473	1
System Introduction	3
Architecture Diagram.....	4
Github Link.....	4
Individual Contribution.....	5
IT22323316 Luvinsion I	5
1. Introduction.....	5
2. Figma Design	6
3. Feedback	8
4. Improvements	8
5. Real User Interface	8
IT22920218 Krushanth A	10
1. Introduction.....	10
2. Figma Design	11
3. Feedback	14
4. Improvements	14
5. Real User Interface	16
IT22346568 Sangeerthan S.....	17
1. Introduction.....	17
2. Github Link - Emergency-Vehicle-Detection-and-Signal-Triggering Github Link	17
3. Figma Design	18
4. Feedback	25
5. Improvements	25
6. Real User Interface	27
IT22000576 Fernandopulle S N.....	30
1. Introduction with system design	30
2. GitHub Link : https://github.com/shriomy/CTDE-MARL-agents.git	31
3. Wireframes and sketches	32
4. Initial Figma Design	33
5. Feedback	34
6. Improvements - Final Figma UI	35

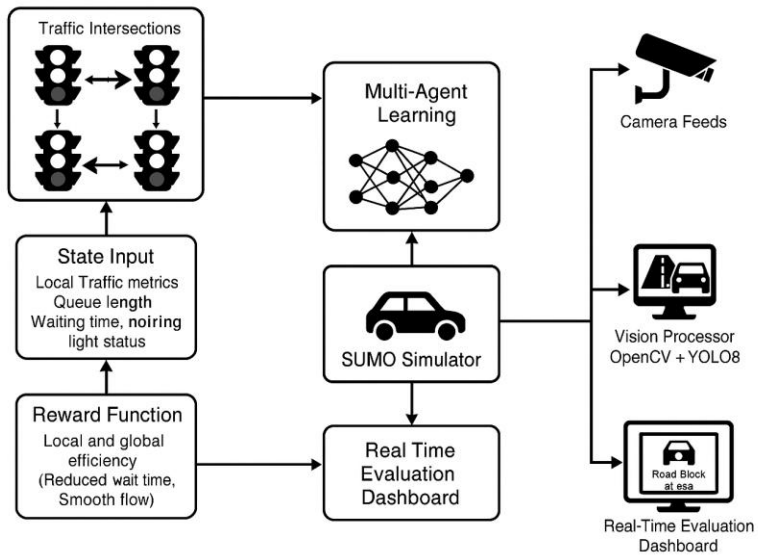
System Introduction

In modern cities, urban traffic congestion is a significant problem that increases travel times, fuel consumption, and safety hazards. Conventional traffic signal systems are unable to adjust to changing traffic conditions, emergency scenarios, and pedestrian demands since they rely on fixed or scheduled time. This research suggests a Smart Traffic Flow Optimization System that uses Multi-Agent Reinforcement Learning (MARL) to get around these restrictions.

In the proposed system, each traffic intersection is modelled as an intelligent agent that observes real-time traffic states such as vehicle queue lengths, waiting times, signal phases, and neighbouring intersection information. To lessen congestion and enhance overall traffic flow, agents learn cooperative signal control policies using a centralized training and decentralized execution (CTDE) strategy using Value Decomposition Networks (VDN). The system combines real-time emergency vehicle detection utilizing audio-based siren recognition and vision-based vehicle identification with an SUMO-based simulation environment. To ensure safer and more equitable urban mobility, an adaptive pedestrian safety module also dynamically modifies crossing signals based on pedestrian behaviour.



Architecture Diagram



Github Link

[System Github Link](#)

Individual Contribution

IT22323316 Luvinsion I

1. Introduction

With the increasing complexity of urban traffic environments, traditional fixed-time traffic signal systems are no longer sufficient to ensure safety and efficiency for all road users. Pedestrians, particularly **students and mobility aid users**, represent vulnerable groups that require additional crossing time and adaptive traffic management strategies. Addressing these needs is essential for the development of inclusive and intelligent transportation systems.

This component is part of the **Smart Traffic Optimisation** project and focuses on **pedestrian detection and classification** to support adaptive pedestrian signal control. The primary objective of the component is to identify pedestrians at crossings and classify them into mobility-based categories, namely **Adult, Elder, Student, and Mobility Aid Users**, using computer vision techniques. The classification results are used to dynamically influence pedestrian signal timings within a simulated traffic environment.

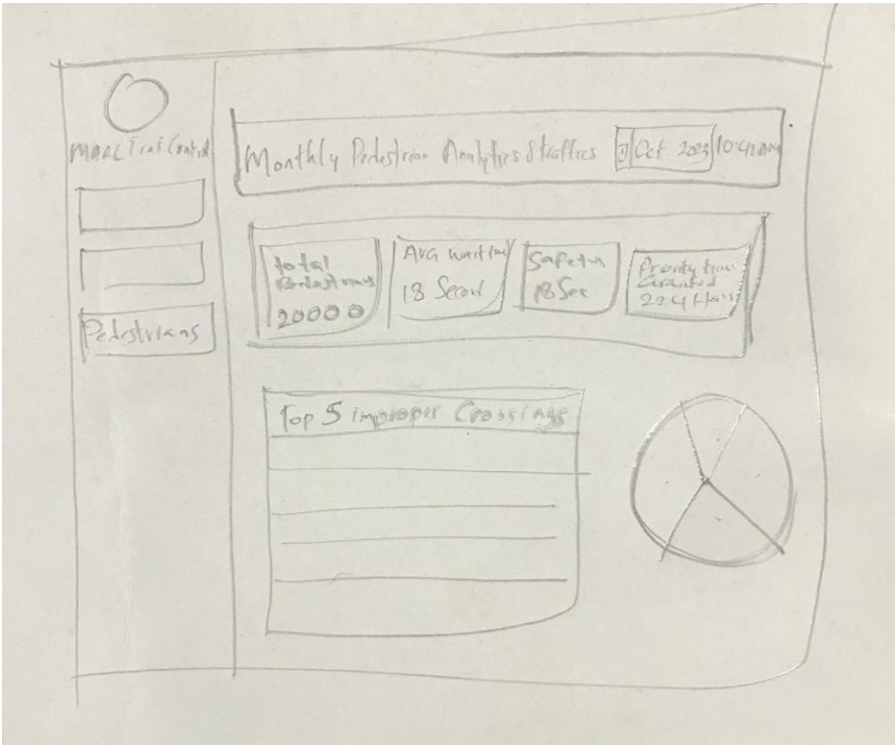
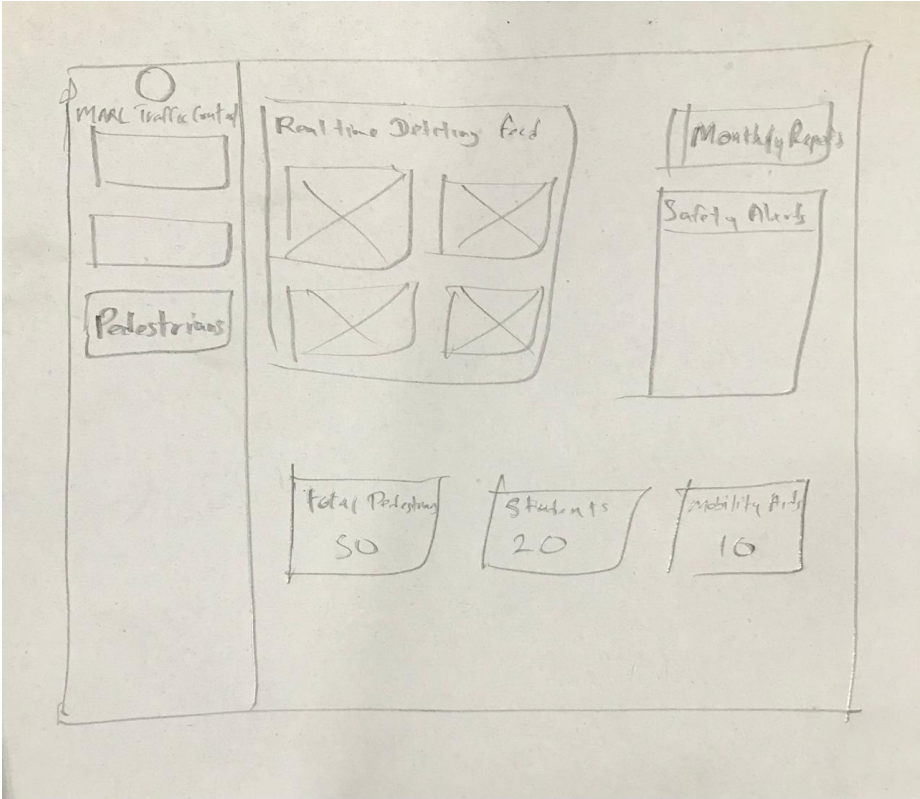
The system employs a deep learning–based object detection model built on the **YOLOv8 architecture**, trained on a custom pedestrian dataset. Visual inputs in the form of images or video streams are processed to detect pedestrians and determine their respective categories. When pedestrians belonging to higher-priority groups, such as students or mobility aid users, are detected, the component signals the traffic control logic to **increase the pedestrian crossing duration**.

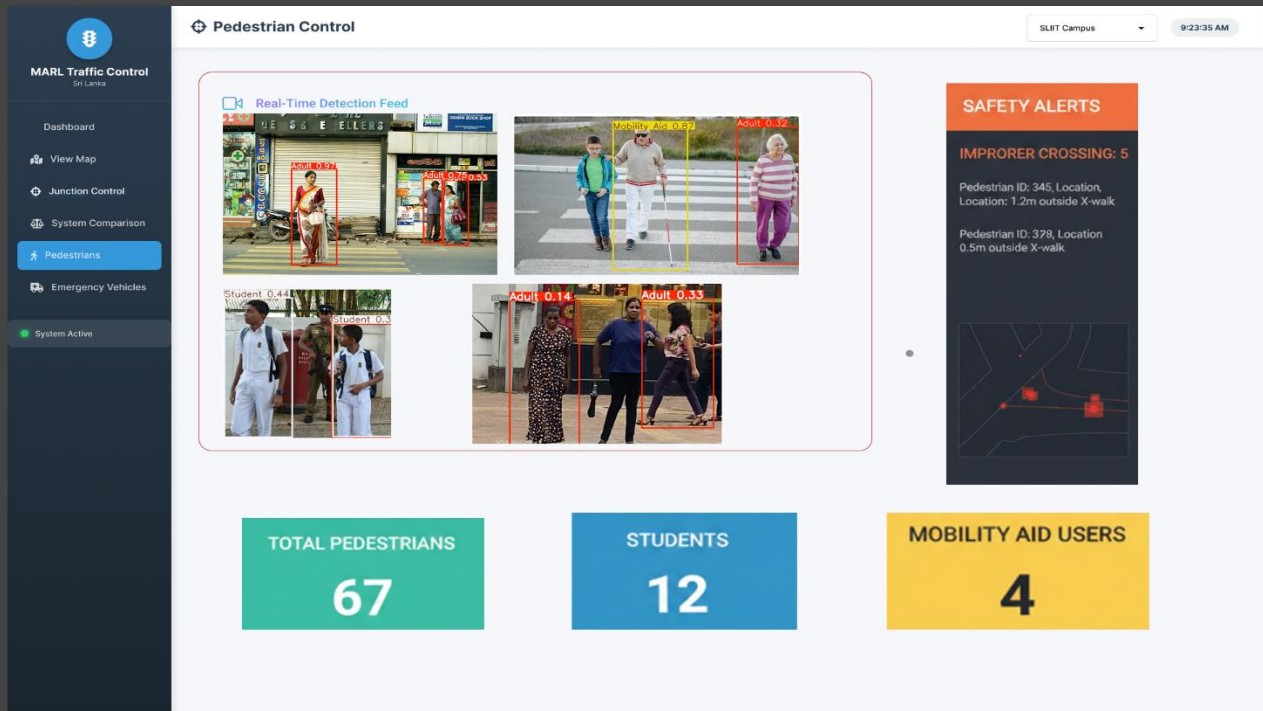
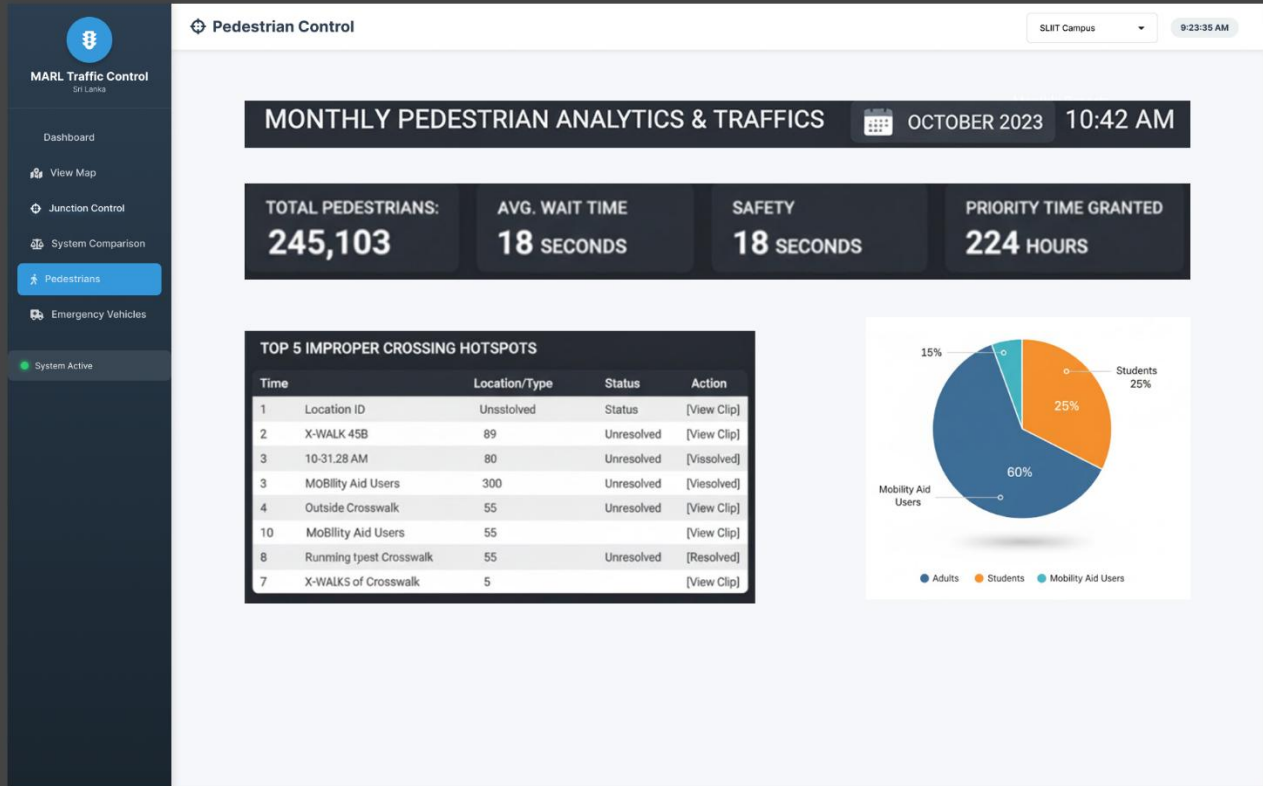
This pedestrian classification output is integrated with **SUMO (Simulation of Urban Mobility)** through a reinforcement learning (RL)–based traffic signal control module developed by another component of the project. The RL agents utilize the pedestrian classification information as part of their state representation, enabling them to make informed decisions on traffic signal phase adjustments. As a result, pedestrian safety is improved while maintaining overall traffic flow efficiency.

By incorporating pedestrian awareness into traffic signal optimization, this component contributes to a more **adaptive, equitable, and intelligent traffic management system**. The proposed approach demonstrates how computer vision and reinforcement learning can be combined to enhance real-world traffic scenarios within a smart city framework.

2. Figma Design

Sketches





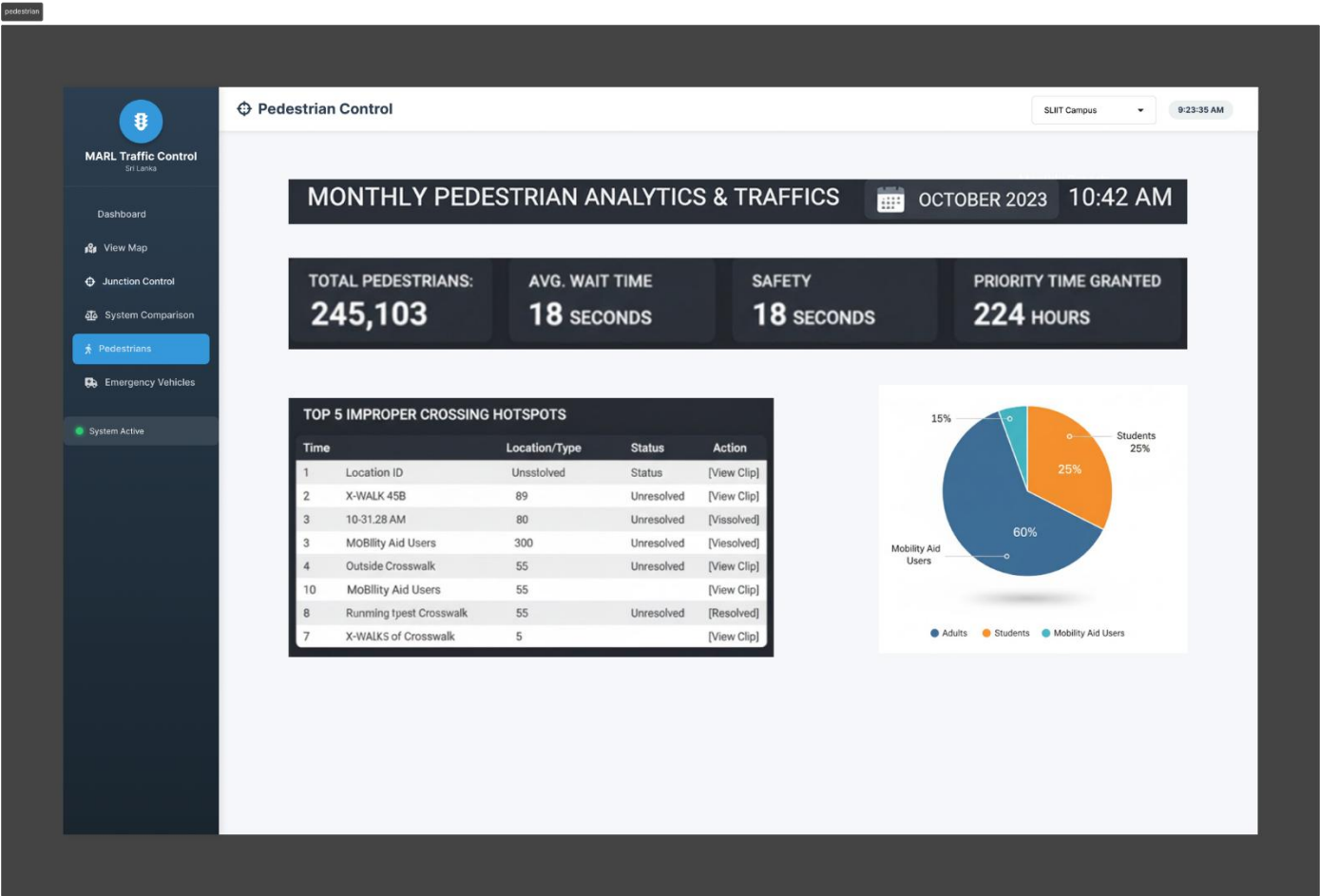
3. Feedback

The pedestrian detection and classification dashboard effectively visualizes pedestrian presence and mobility categories, enabling adaptive pedestrian signal timing within the Smart Traffic Optimisation system. Its integration with reinforcement learning–based traffic signal agents allows the system to prioritize vulnerable pedestrians such as students and mobility aid users, thereby improving safety and traffic efficiency in simulated urban environments.

4. Improvements

To further enhance system capability, an additional requirement is the detection and counting of pedestrians who do not follow traffic rules. This includes identifying pedestrians crossing outside designated crosswalks or attempting to cross during red signal phases. Incorporating violation detection and maintaining counts of such unsafe behaviors would provide valuable feedback to the dashboard and reinforcement learning agents, enabling safety-aware signal control decisions and supporting more comprehensive traffic behavior analysis.

5. Real User Interface





MARL Traffic Control
SRI Lanka

Dashboard

View Map

Junction Control

System Comparison

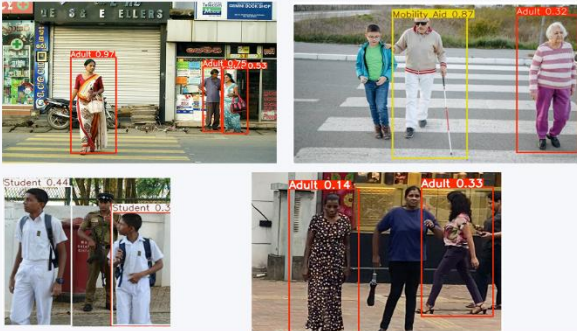
Pedestrians

Emergency Vehicles

System Active

Pedestrian Control

Real-Time Detection Feed



TOTAL PEDESTRIANS

67

STUDENTS

12

MOBILITY AID USERS

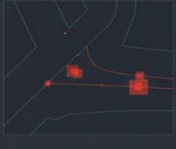
4

SAFETY ALERTS

IMPROPER CROSSING: 5

Pedestrian ID: 345, Location:
Location: 1.2m outside X-walk

Pedestrian ID: 379, Location:
Location: 0.5m outside X-walk



1. Introduction

With the growing complexity of urban road networks and the rapid increase in vehicle density, traditional rule-based and fixed-time traffic signal control methods are no longer adequate to ensure efficient traffic flow. Static signal timings fail to respond to real-time traffic conditions, leading to congestion, increased travel time, and inefficient utilization of road infrastructure. To address these challenges, intelligent and adaptive traffic control mechanisms are essential components of modern smart transportation systems. This component is part of the Smart Traffic Flow Optimization project and focuses on optimizing traffic signal timing using a Single-Agent Reinforcement Learning (RL) approach. The primary objective of this component is to develop an intelligent traffic signal controller capable of learning optimal signal phase decisions based on real-time traffic conditions at an intersection.

By continuously interacting with the traffic environment, the RL agent aims to minimize congestion, reduce vehicle waiting time, and improve overall traffic efficiency. The system models the traffic signal controller as a single learning agent operating within a simulated traffic environment. The agent observes the current state of the intersection, which includes traffic density, queue lengths, and signal phase information, and selects appropriate actions such as extending or switching traffic signal phases. Through a reward-based learning process, the agent gradually learns policies that balance traffic flow across all approaches of the intersection.

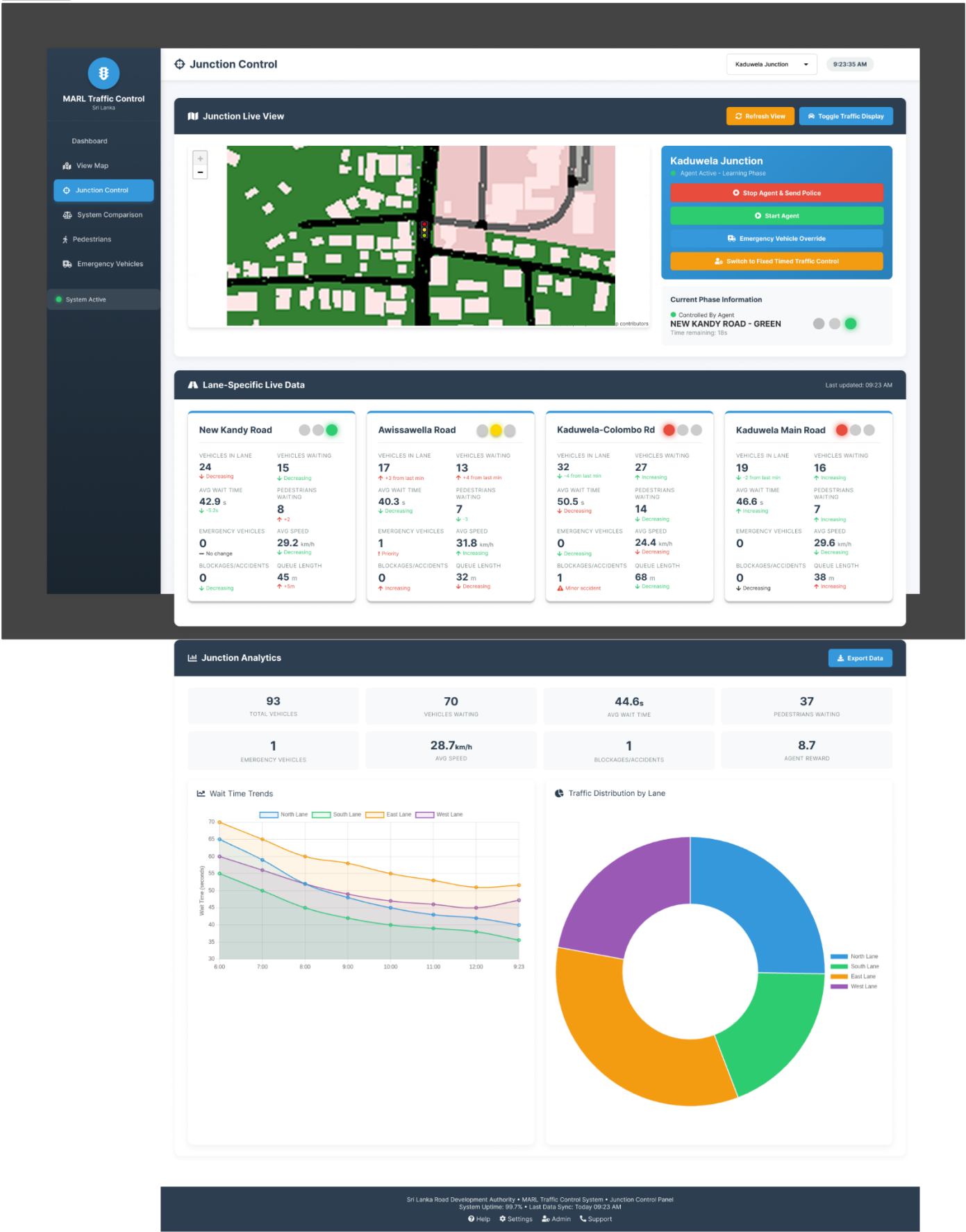
The traffic environment is simulated using SUMO (Simulation of Urban Mobility), allowing realistic modeling of vehicle movements, traffic demand, and intersection behavior. Reinforcement learning algorithms are trained through repeated simulation episodes, enabling the agent to adapt to varying traffic patterns without relying on predefined rules or fixed schedules. This single-agent reinforcement learning approach provides a scalable and effective solution for intelligent traffic signal control. By enabling autonomous decision-making based on real-time traffic states, this component contributes to a more adaptive, efficient, and intelligent traffic management system. The outcomes of this component demonstrate the potential of reinforcement learning techniques to improve urban traffic flow within a smart city framework.

Github Link

<https://github.com/KrushanthAmalanathan/RP-25-26J-473-IT22920218>

2. Figma Design

junction control - kaduwela





MARL Traffic Control
Sri Lanka

Dashboard

View Map

Junction Control

System Comparison

Pedestrians

Emergency Vehicles

System Active

Junction Control

SLIT Campus 9:23:35 AM

Junction Live View

Refresh View Toggle Traffic Display



SLIT Campus - Malabe

Agent Active - Learning Phase

Stop Agent & Send Police

Start Agent

Emergency Vehicle Override

Switch to Fixed Timed Traffic Control

Current Phase Information

Controlled By Fixed Timed Traffic Controller

KADUWELA ROAD NORTH - GREEN

Time remaining: 38s

Lane-Specific Live Data

Last updated: 09:23 AM

New Kandy Road

VEHICLES IN LANE
24
Decreasing

VEHICLES WAITING
15
Decreasing

AVG WAIT TIME
42.9 s
Decreasing

PEDESTRIANS WAITING
8
Increasing

EMERGENCY VEHICLES
0
No change

AVG SPEED
29.2 km/h
Decreasing

BLOCKAGES/ACCIDENTS
0
Decreasing

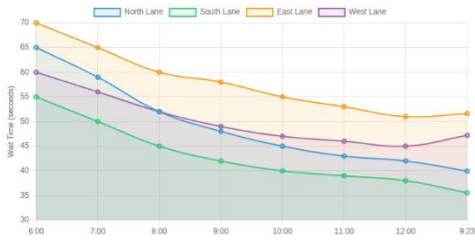
QUEUE LENGTH
45 m
Increasing

Junction Analytics

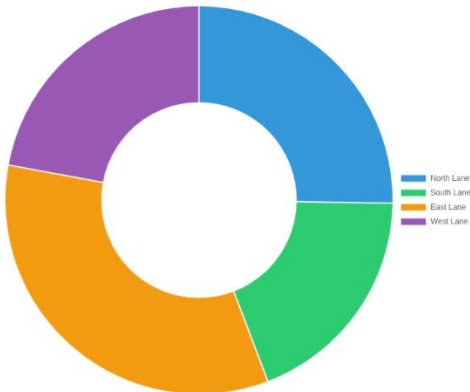
Export Data

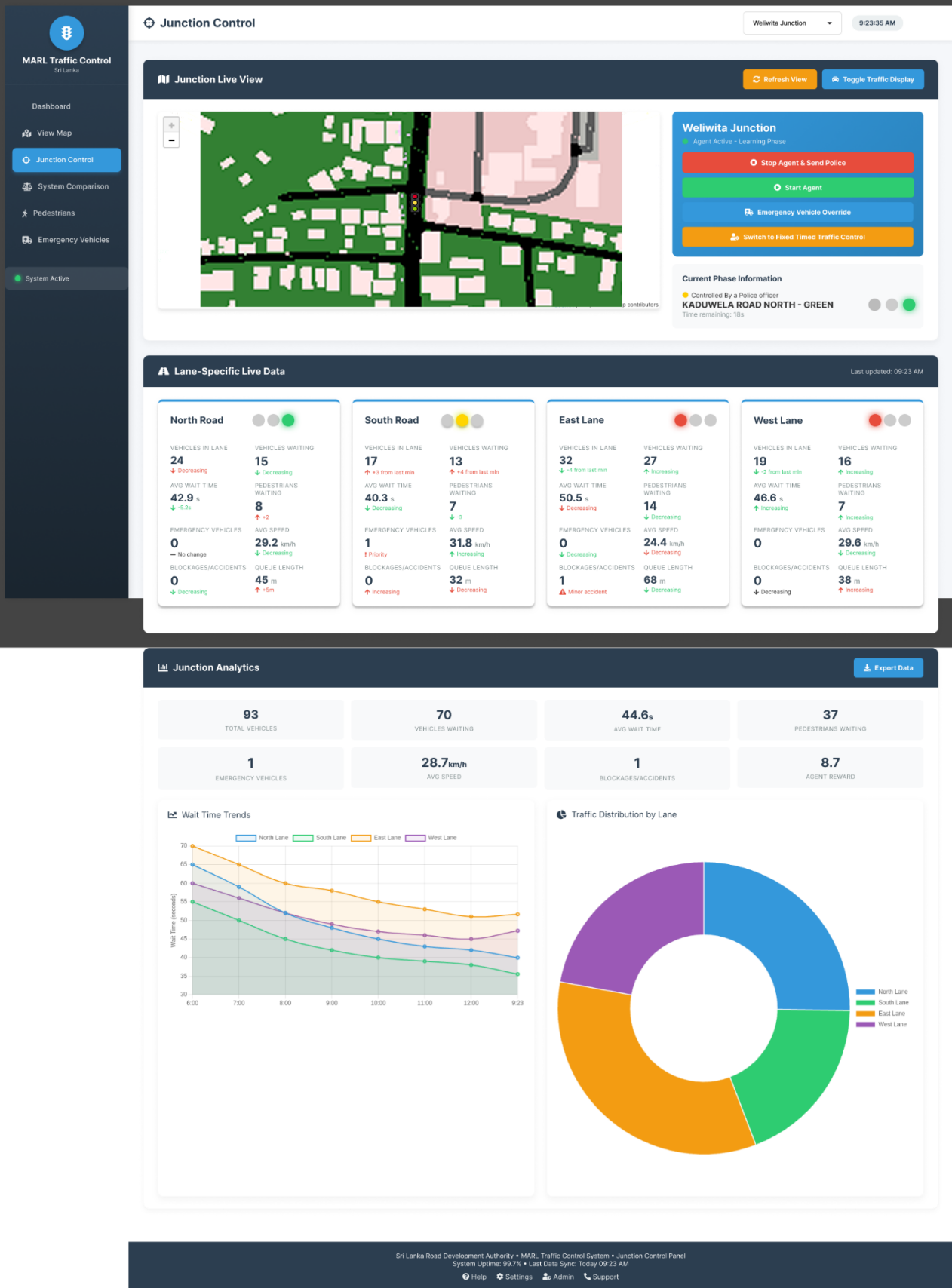
93 TOTAL VEHICLES	70 VEHICLES WAITING	44.6s AVG WAIT TIME	37 PEDESTRIANS WAITING
1 EMERGENCY VEHICLES	28.7km/h AVG SPEED	1 BLOCKAGES/ACCIDENTS	8.7 AGENT REWARD

Wait Time Trends



Traffic Distribution by Lane





3. Feedback

Based on feedback from users reviewing the initial design, the following issues were identified:

- The interface appeared **too complex** for quick understanding
- Important information such as **current green signal and queue priority** was not immediately noticeable
- Decision explanations were **not clear enough** for non-technical users
- Excessive data on a single screen caused **cognitive overload**
- Manual override and system control options were not clearly separated from monitoring elements

This feedback highlighted the need for a cleaner layout, clearer decision justification, and better prioritization of information.

4. Improvements

Structural Improvements

- Simplified dashboard layout with **clear section separation**
- Improved spacing and alignment to reduce visual clutter
- Logical grouping of related information (junction overview, signal status, queues)

Usability Enhancements

- Clear display of **current active signal direction** with remaining time
- Reduced number of metrics shown at once to focus on decision-critical data
- More readable queue score presentation per direction

Decision Transparency Improvements

- Added a dedicated **Decision Reason** section to clearly explain why a signal is being held or switched
- Simplified decision labels (e.g., “holding north”) for better user understanding

Wireframes

Intersection SummaryView

↑

NORTH

2

↓

SOUTH

1

→

EAST

0

←

WEST

3

Sensors ConnectivitySOUTH

NorthConnected

SouthConnected

EastDisconnected

WestConnected

North - Traffic Data	South - Traffic Data	East - Traffic Data	West - Traffic Data
Queue Length12 vehicles	Queue Length8 vehicles	Queue Length15 vehicles	Queue Length10 vehicles
Wait Time45s	Wait Time32s	Wait Time58s	Wait Time40s
Green Duration60s	Green Duration55s	Green Duration65s	Green Duration58s
Red Duration90s	Red Duration85s	Red Duration95s	Red Duration88s
Flow Rate240 veh/hr	Flow Rate200 veh/hr	Flow Rate280 veh/hr	Flow Rate220 veh/hr
View Details	View Details	View Details	View Details

Special Override

RED

Duration: 30s

YELLOW

Duration: 5s

Counter Overview

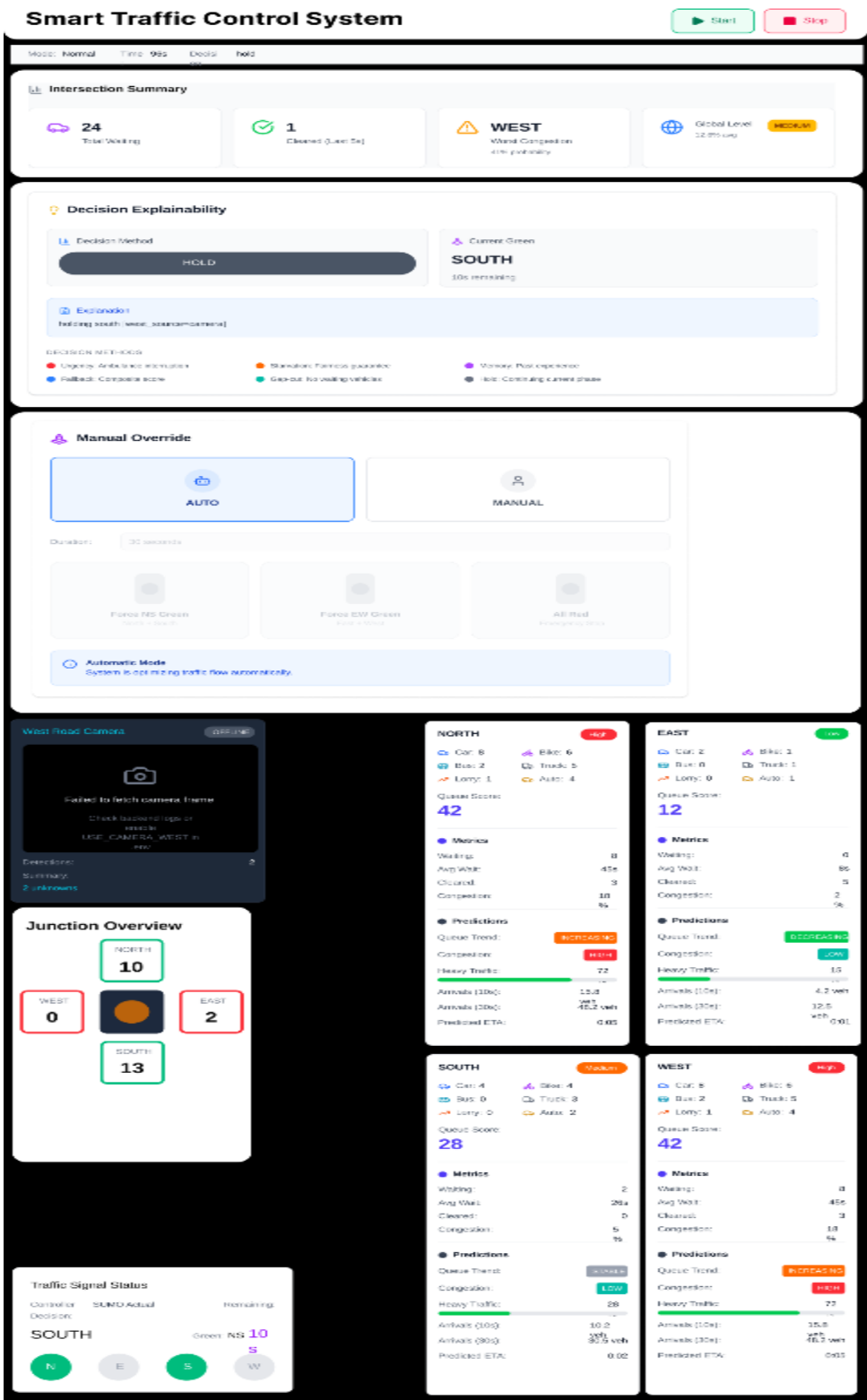
10

5

15

StartPauseReset

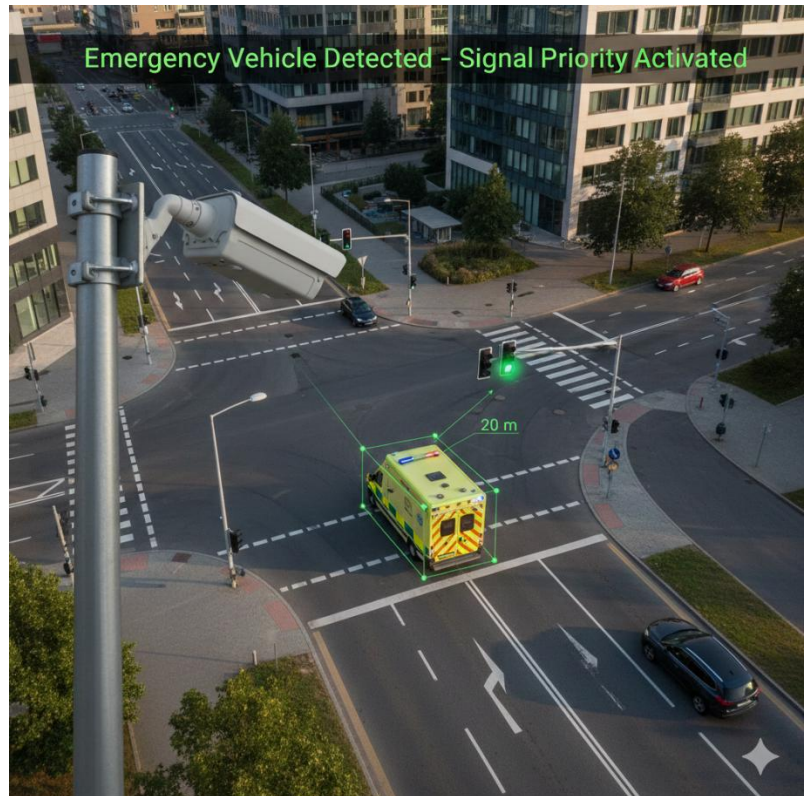
5. Real User Interface



1. Introduction

Rapid identification of emergency vehicles is a critical requirement in modern smart city traffic management systems. Delays in recognizing ambulances, fire trucks, and police vehicles can lead to increased response times, traffic congestion, and potential loss of life.

This AI-based Emergency Vehicle Detection System presented in this research is intended to precisely detect emergency vehicles in real-time for intelligent traffic management applications. To provide intelligent traffic management, the system uses YAMNet for siren sound identification and YOLOv11 for front-view and real-time top-view vehicle detection. Because the top-view camera is placed close to traffic signals, it can accurately identify emergency vehicles like police cars, fire engines, and ambulances by reducing occlusion in heavy traffic. The technology supports proactive traffic signal control and quicker emergency response by combining visual detection with audio-based siren recognition to provide reliable identification even in low visibility or high traffic

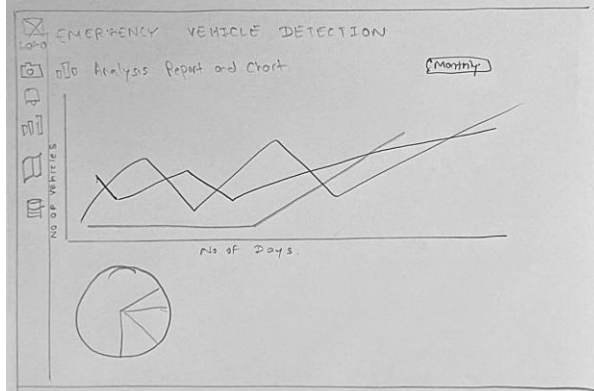
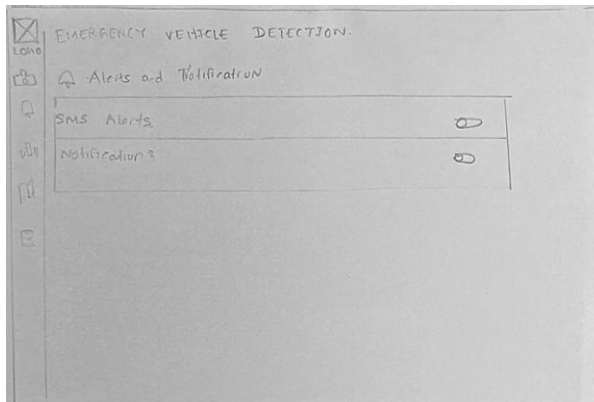
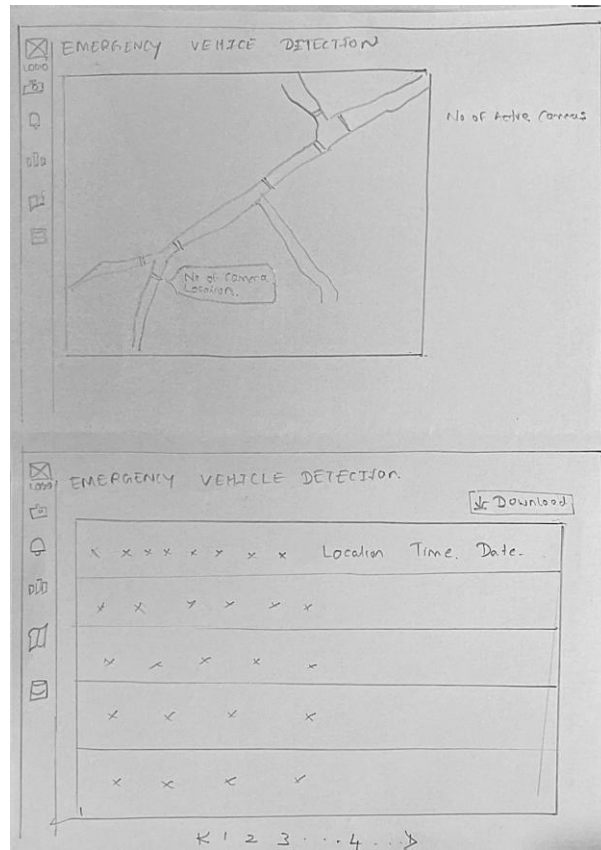
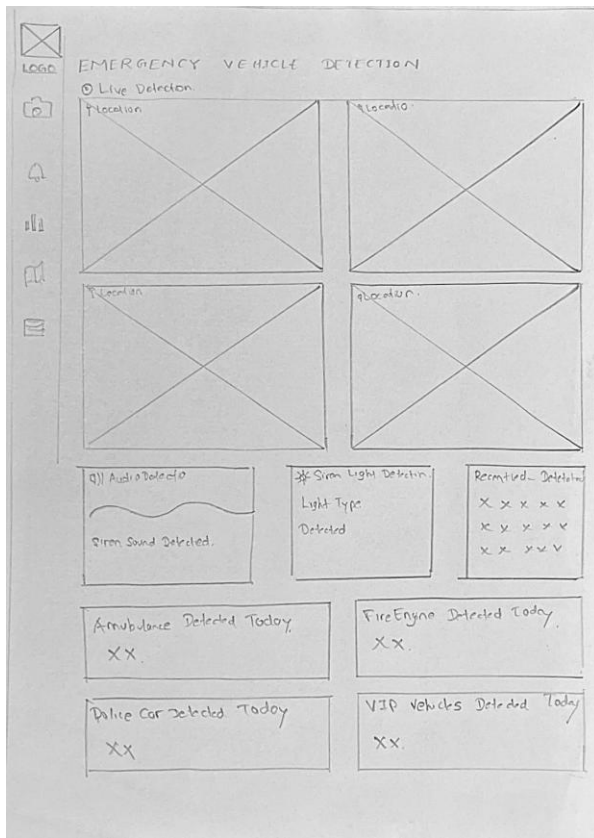


It recognizes emergency-related text ("AMBULANCE", "FIRE AND RESCUE" or "POLICE") and recognizes common emergency symbols on cars by combining computer vision techniques with EasyOCR. Additionally, the system uses audio signal processing to identify siren sounds and flashing light pattern analysis to recognize siren lights in order to increase dependability. The suggested solution guarantees reliable and precise emergency vehicle recognition by integrating optical, textual, and audio-based detection techniques. This allows for quicker reaction times, automated traffic prioritization, and enhanced road safety in smart city settings.

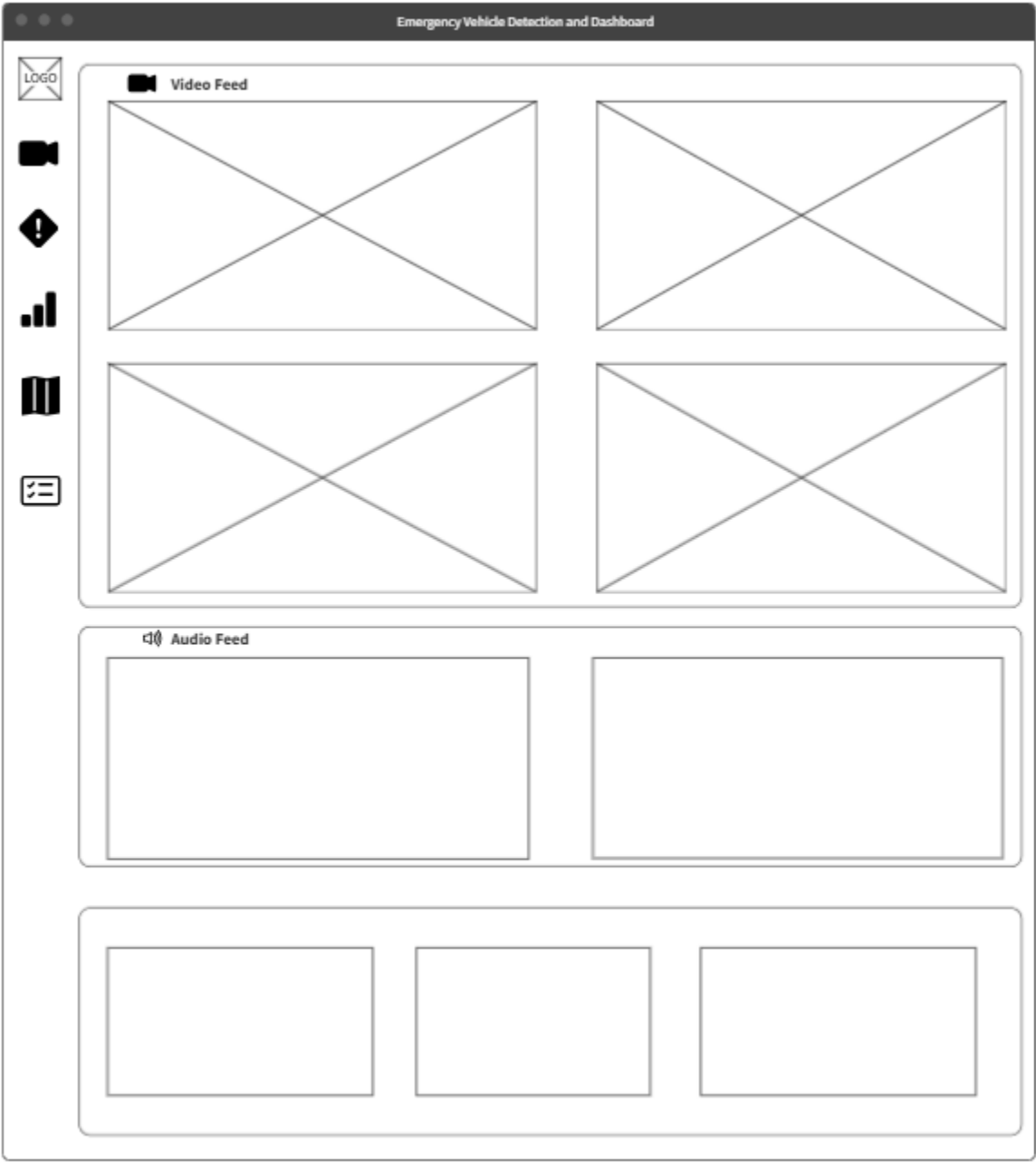
2. Github Link - [Emergency-Vehicle-Detection-and-Signal-Triggering Github Link](#)

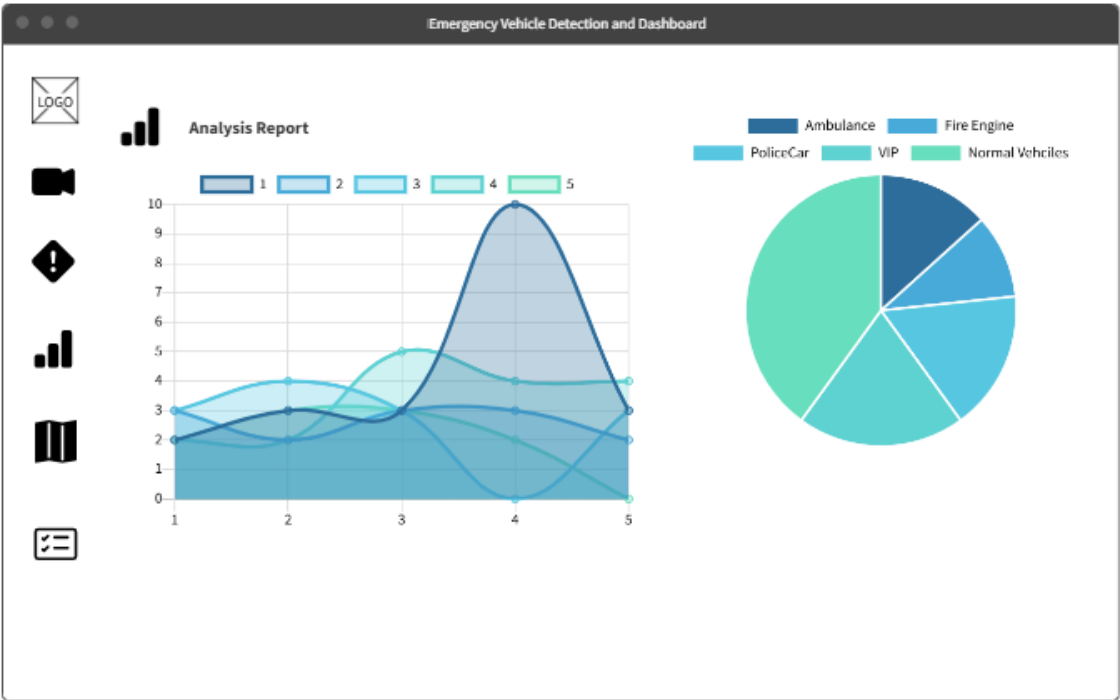
3. Figma Design

Sketches :



Wireframes:



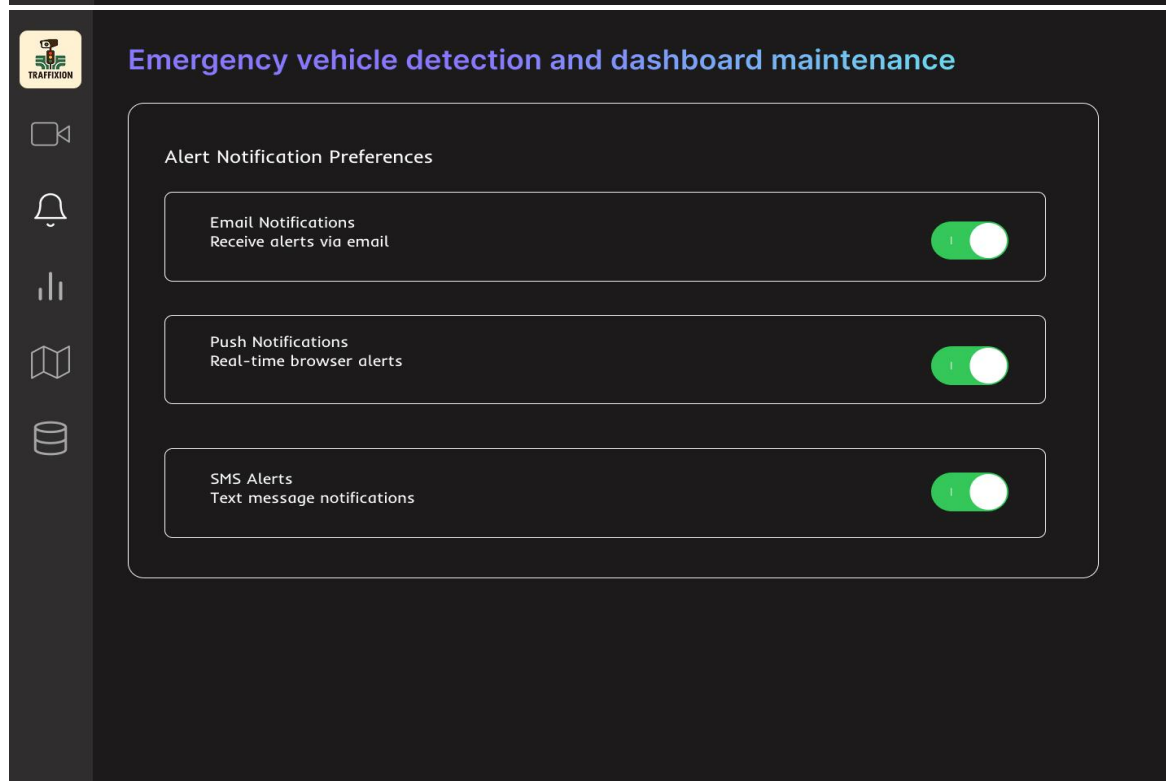
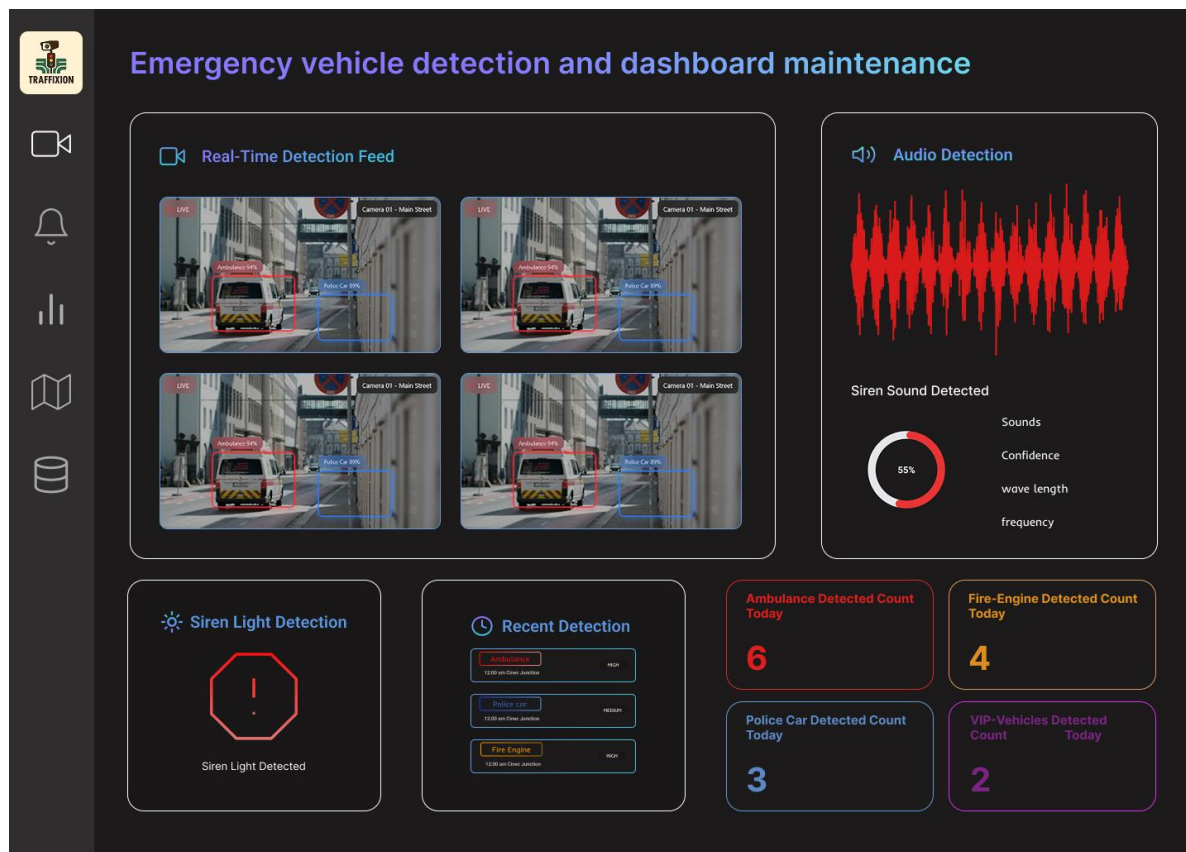


Emergency Vehicle Detection Management

☐	EV	Location
☒	Ambulance	Weliwita
☐	Fire Engine	CINEC
☒	Police Car	SPAR
☐	VIP	Kaduwela
☒	Ambulance	Weliwita
☐	Fire Engine	CINEC
☒	Police Car	SPAR

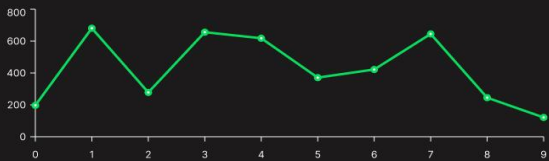
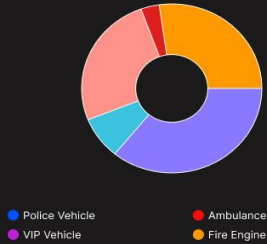
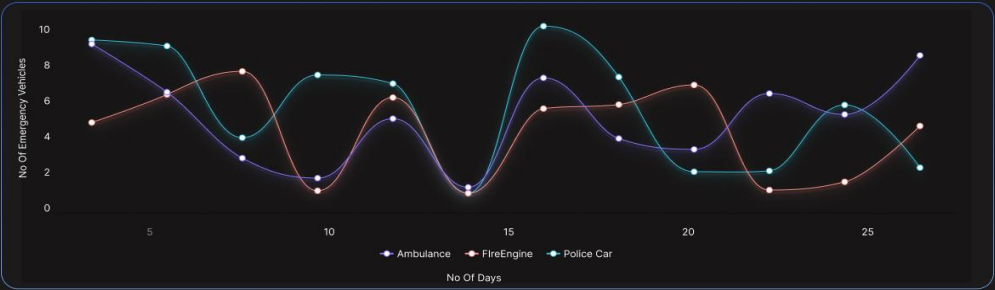
« 1 2 3 4 5 »







Emergency vehicle detection and dashboard maintenance



Emergency vehicle detection and dashboard maintenance

Alerts & Detection Logs

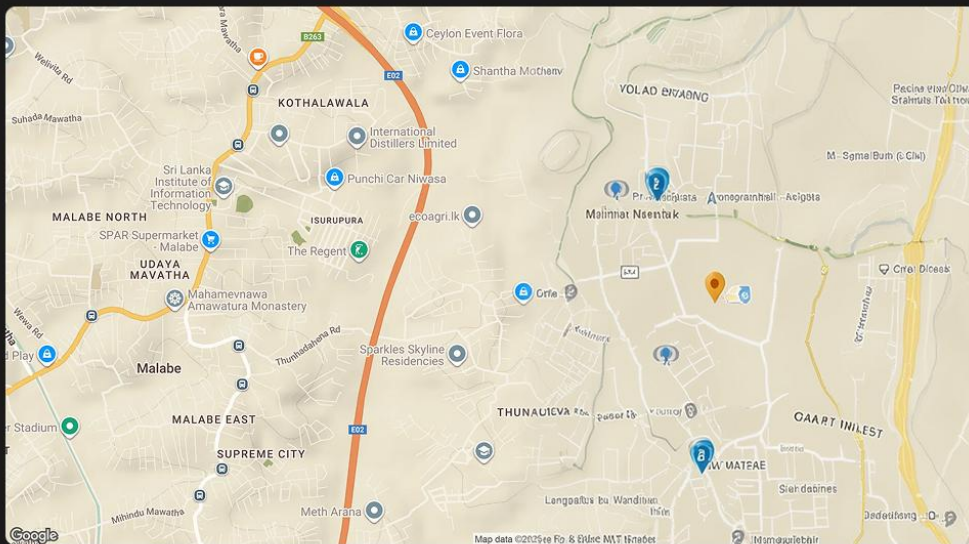
Q Search item by Vehicle Name	All Vehicles ▼	📄 Export CSV				
Detection ID	Vehicle Type	Siren	Flashing Light	Confidence	Timestamp	Source
DET-2024-007	Fire Engine	Yes	Yes	95%	2024-12-09 13:45:21	Camera 02 - Hospital
DET-2024-008	Ambulance	No	Yes	88%	2024-12-09 13:30:05	Camera 01 - Main St
DET-2024-009	Police Car	Yes	No	90%	2024-12-09 13:25:48	Mic 03 - Downtown
DET-2024-010	VIP Vehicle	No	Yes	93%	2024-12-09 13:10:12	Camera 04 - Plaza
DET-2024-011	Ambulance	Yes	Yes	97%	2024-12-09 12:55:30	Mic 01 - Oak St
DET-2024-012	Fire Engine	Yes	No	86%	2024-12-09 12:40:55	Camera 03 - Downtown



Emergency vehicle detection and dashboard maintenance



 Map Location



-  Active Camera
-  Active Mic
-  Alert

4. Feedback

Users reported that the emergency vehicle detection system was easy to understand and intuitive to use, mentioning location-based UI.

The real-time visualization of detected emergency vehicles was considered clear and informative.

The dashboard clearly highlights detected emergency vehicles using bold colors, icons, and alerts, allowing me to quickly understand the vehicle type, confidence level, and recommended signal action without reading complex data.

5. Improvements

Improved the visual alerts, colors, and icons so emergency vehicles are now identified instantly, helping users recognize critical situations within seconds.

Added confidence indicators and detection details (vehicle type, OCR text, siren status), making it easier for users to understand and trust why the system takes a specific traffic control action.

Initial UI was two page to see the live feed and get the response , the user asked to keep them all in one desktop page to easy usage. -

Emergency vehicle detection and dashboard maintenance



Real-Time Detection Feed



Audio Detection



Siren Sound Detected



Sounds
Confidence
frequency
wave length

Siren Light Detection



Siren Light Detected

Recent Detection

Ambulance	HIGH
12:00 am Clisec Junction	
Police car	MEDIUM
12:00 am Clisec Junction	
Fire Engine	HIGH
12:00 am Clisec Junction	

Ambulance Detected Count
Today

6

Fire-Engine Detected Count
Today

4

Police Car Detected Count
Today

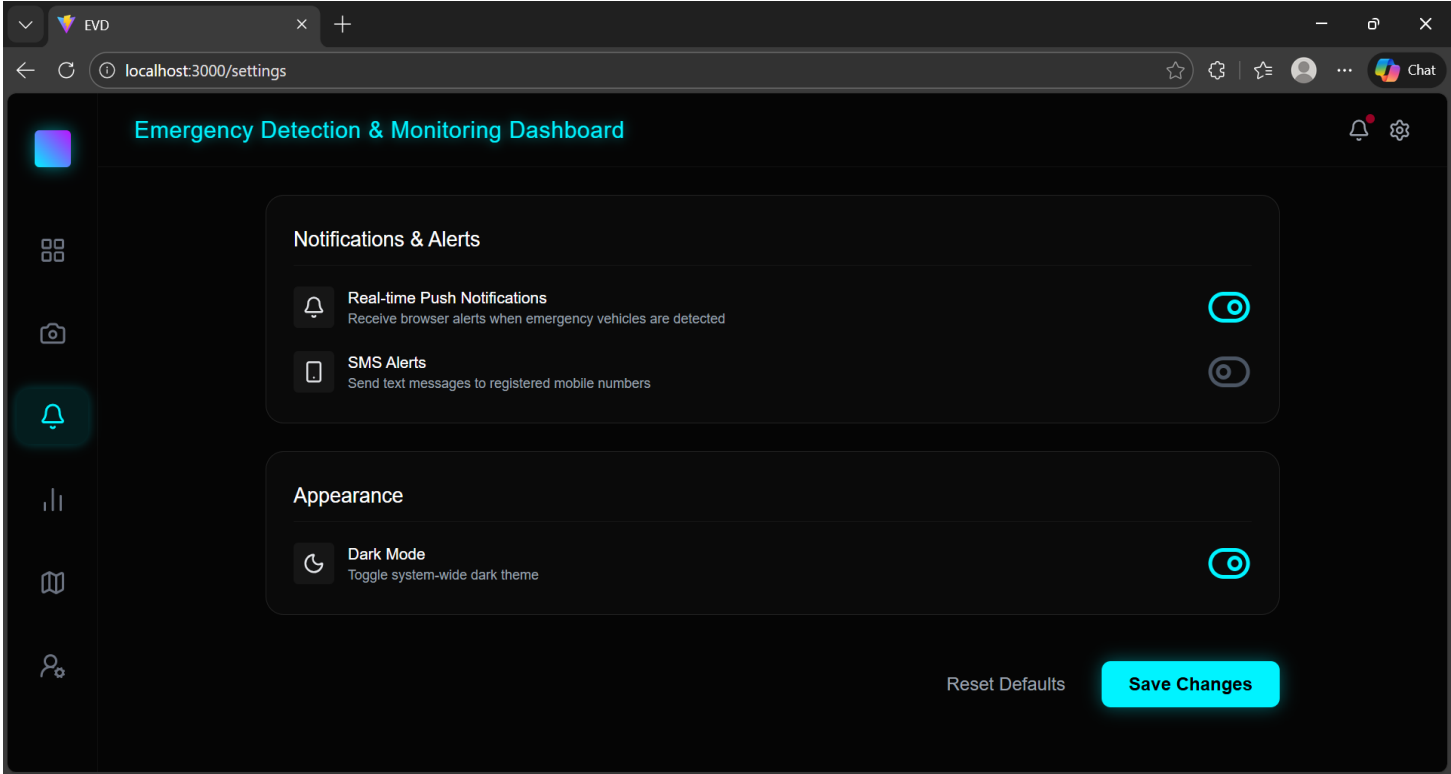
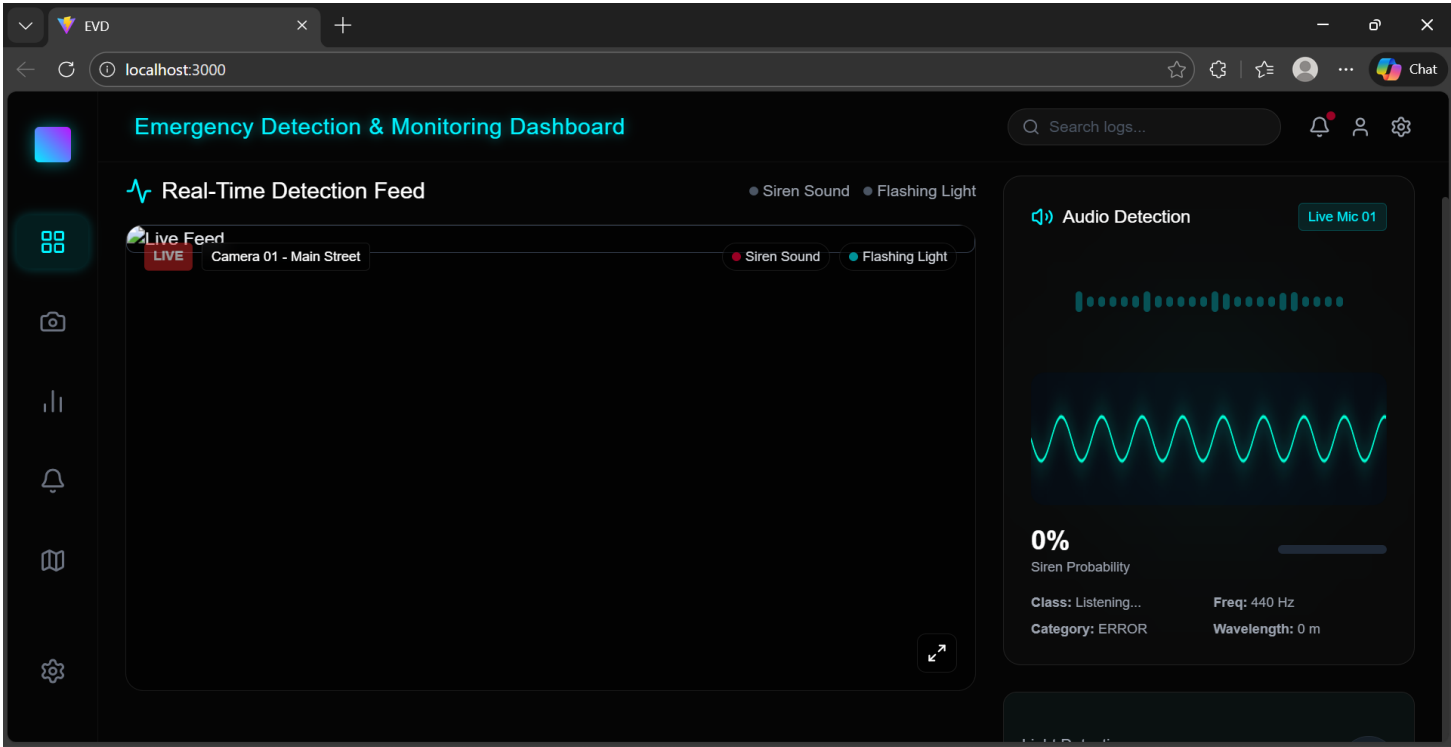
3

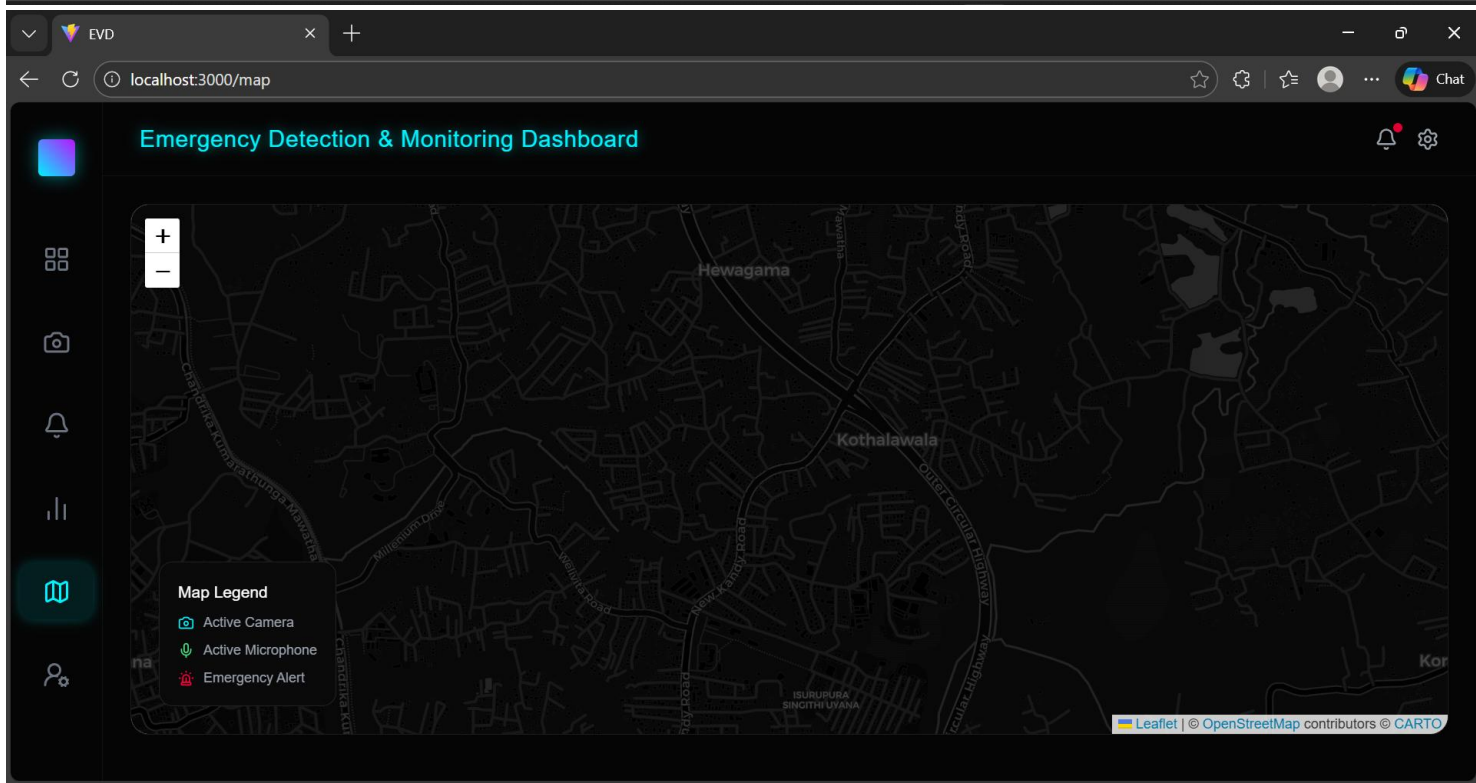
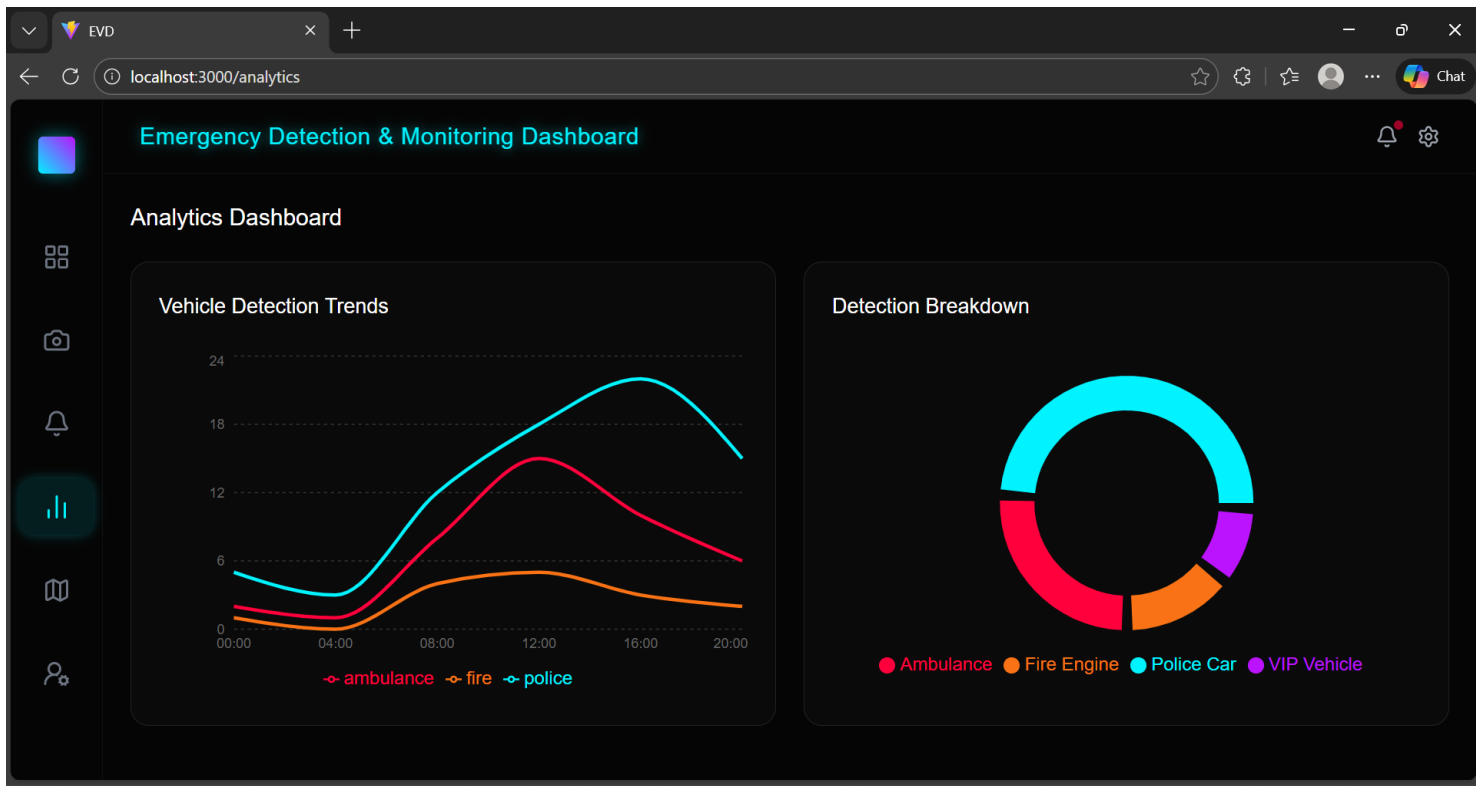
VIP-Vehicles Detected Count
Today

2



6. Real User Interface





localhost:3000/alerts
⌵ ⌶ ⌷ Chat

Emergency Detection & Monitoring Dashboard

Alerts & System Logs

Export

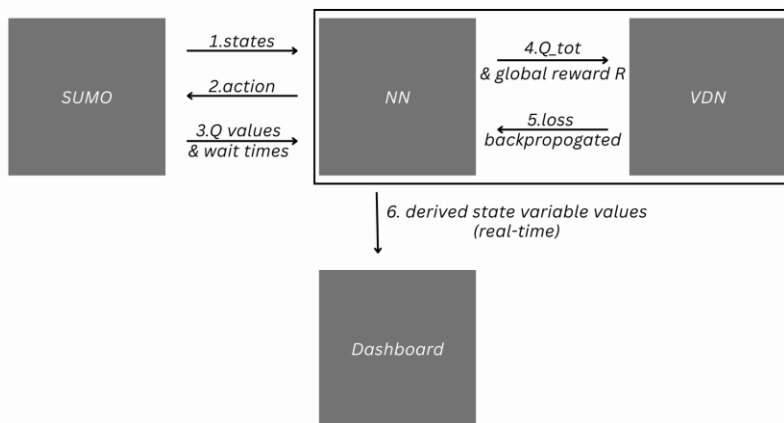
Status	Detection Type	Timestamp	Location	Confidence	Actions
Critical	Ambulance	2 min ago	Main St / 4th Ave	94%	View Details
Warning	Police Car	5 min ago	Broadway End	89%	View Details
Critical	Fire Engine	12 min ago	Industrial Zone	91%	View Details
Critical	VIP Vehicle	25 min ago	City Center	98%	View Details
Warning	Ambulance	40 min ago	Highway Exit 5	88%	View Details
Info	System Update	Just now	All Systems	100%	View Details
Info	Network Status	Just now	All Nodes	100%	View Details

1. Introduction with system design

The proposed system is an intelligent traffic signal control platform based on Multi-Agent Reinforcement Learning (MARL) using the Value Decomposition Network (VDN) framework under the Centralized Training and Decentralized Execution (CTDE) paradigm. It is designed to manage and coordinate multiple traffic intersections simultaneously with the goal of minimizing overall congestion, travel time, and network-wide delay in urban road networks.

The system treats each traffic intersection as an autonomous learning agent that controls signal phases based on real-time traffic conditions. Rather than optimizing intersections in isolation, the system enables cooperative decision-making across the entire road network. During centralized training, a global learning process observes the joint state of all intersections and evaluates a shared reward that reflects overall traffic efficiency. VDN decomposes this global value function into individual agent value functions, allowing each intersection to understand how its local actions contribute to global performance.

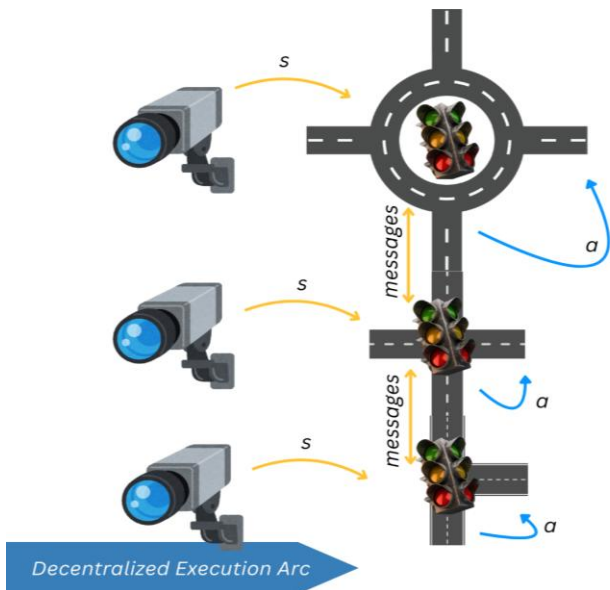
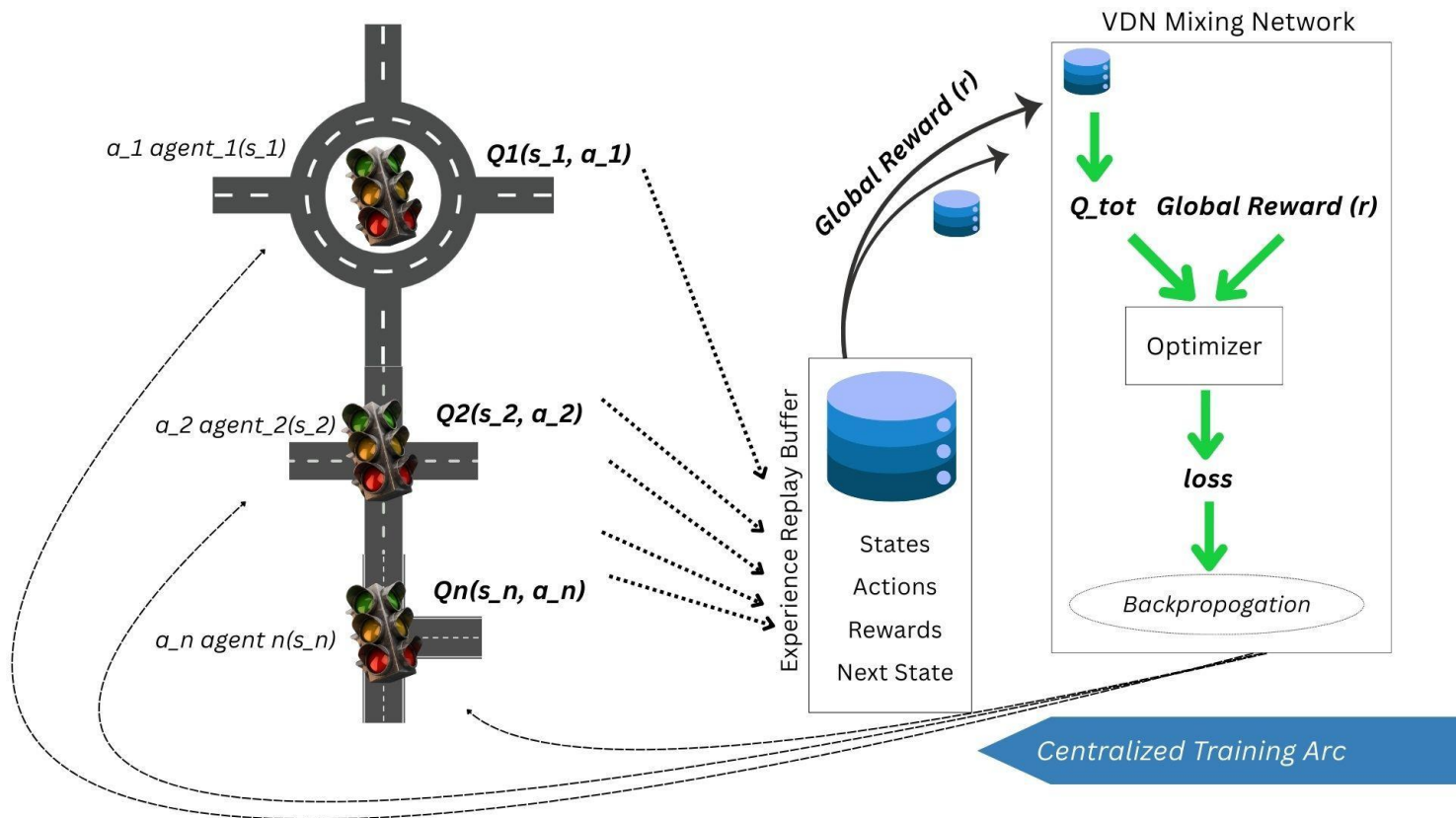
This is a simplified diagram to the system architecture on how it will be integrated.



21

Once training is complete, execution is fully decentralized. Each intersection operates independently using its learned policy, relying only on local observations and lightweight communication with neighboring intersections. This makes the system scalable, robust, and suitable for real-time deployment. Integrated with the SUMO traffic simulator via TraCI, the system adapts dynamically to changing traffic patterns, supports coordinated behaviors such as green waves, and provides a strong foundation for integrating emergency vehicle prioritization and pedestrian-aware control.

The diagram on how the centralized training works for the MARL agents using VDN architecture.



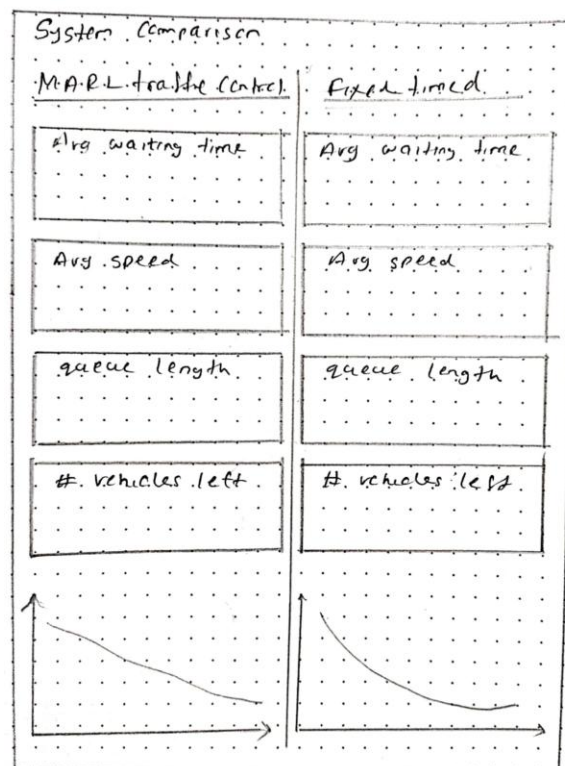
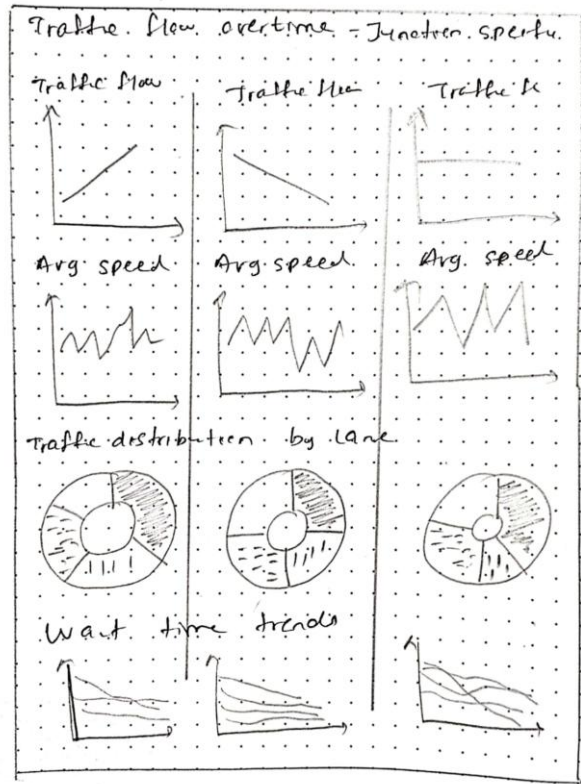
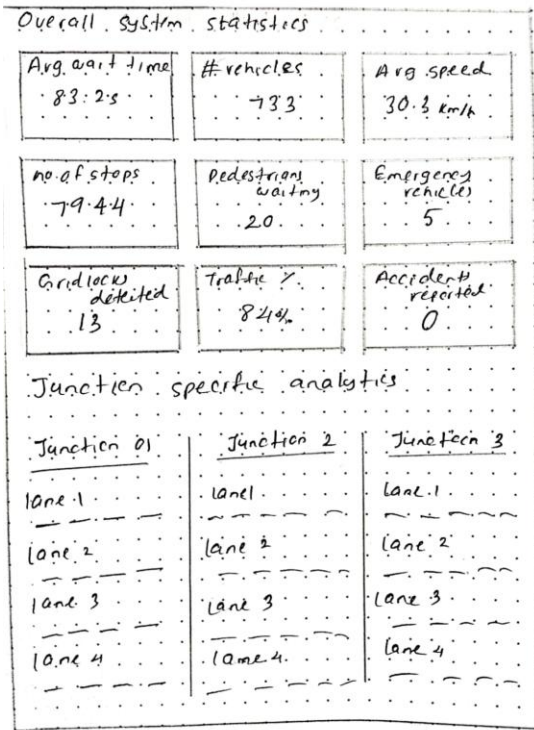
The diagram for decentralized execution in the MARL environment. This approach reduces traffic congestion by enabling coordinated decision-making among intersections. Through centralized training and decentralized execution, agents learn cooperative signal strategies that balance traffic flow across the network, prevent bottleneck shifting, adapt to dynamic conditions, and minimize overall vehicle waiting time and travel delays.

2. GitHub Link : <https://github.com/shriomy/CTDE-MARL-agents.git>

3. Wireframes and sketches

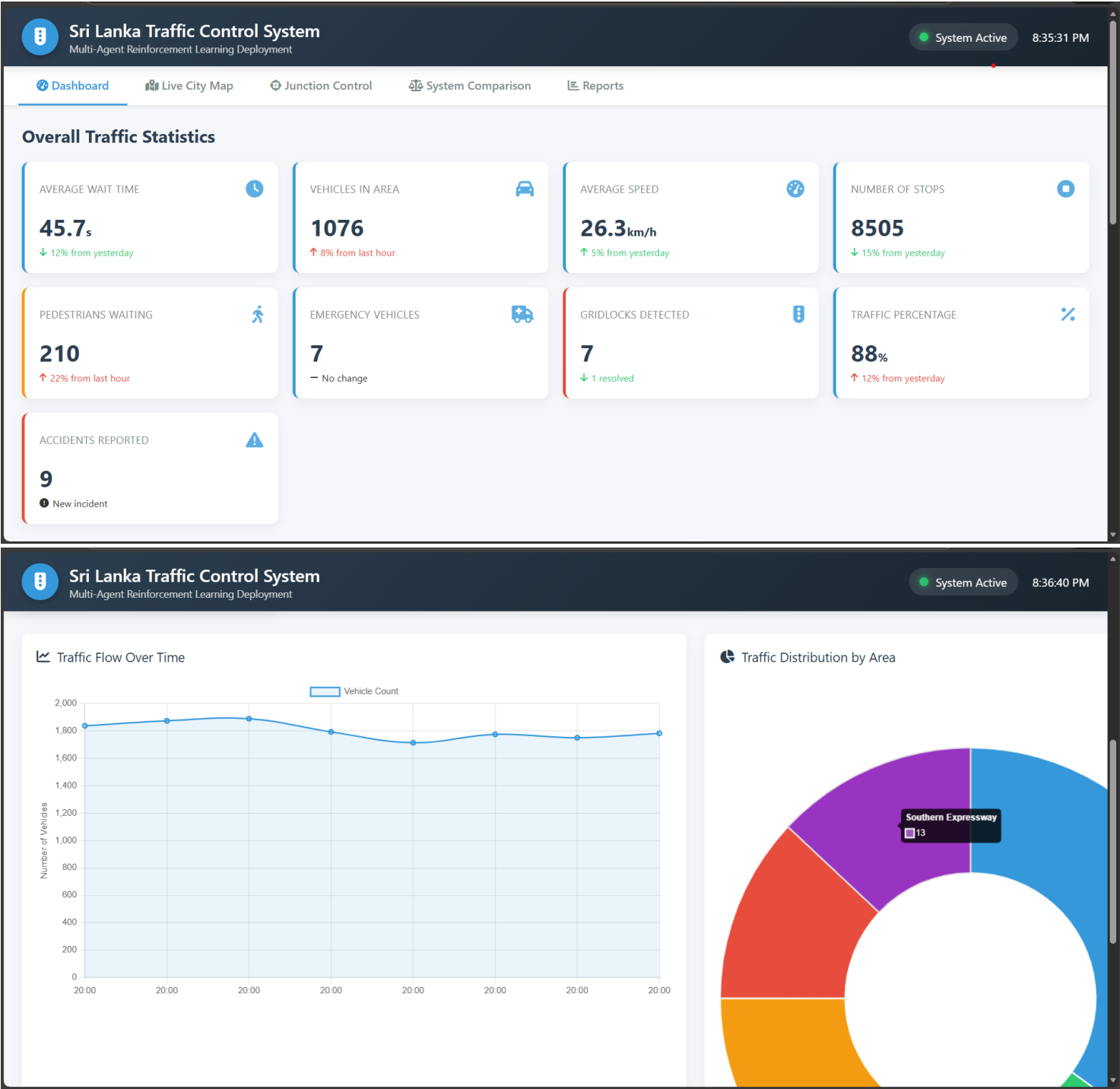
The user interfaces related to this specific components are

1. **dashboard** to collect city wide traffic statistics and a,
2. **traffic prediction and comparison** between different environment systems for flexibility to switch between them.

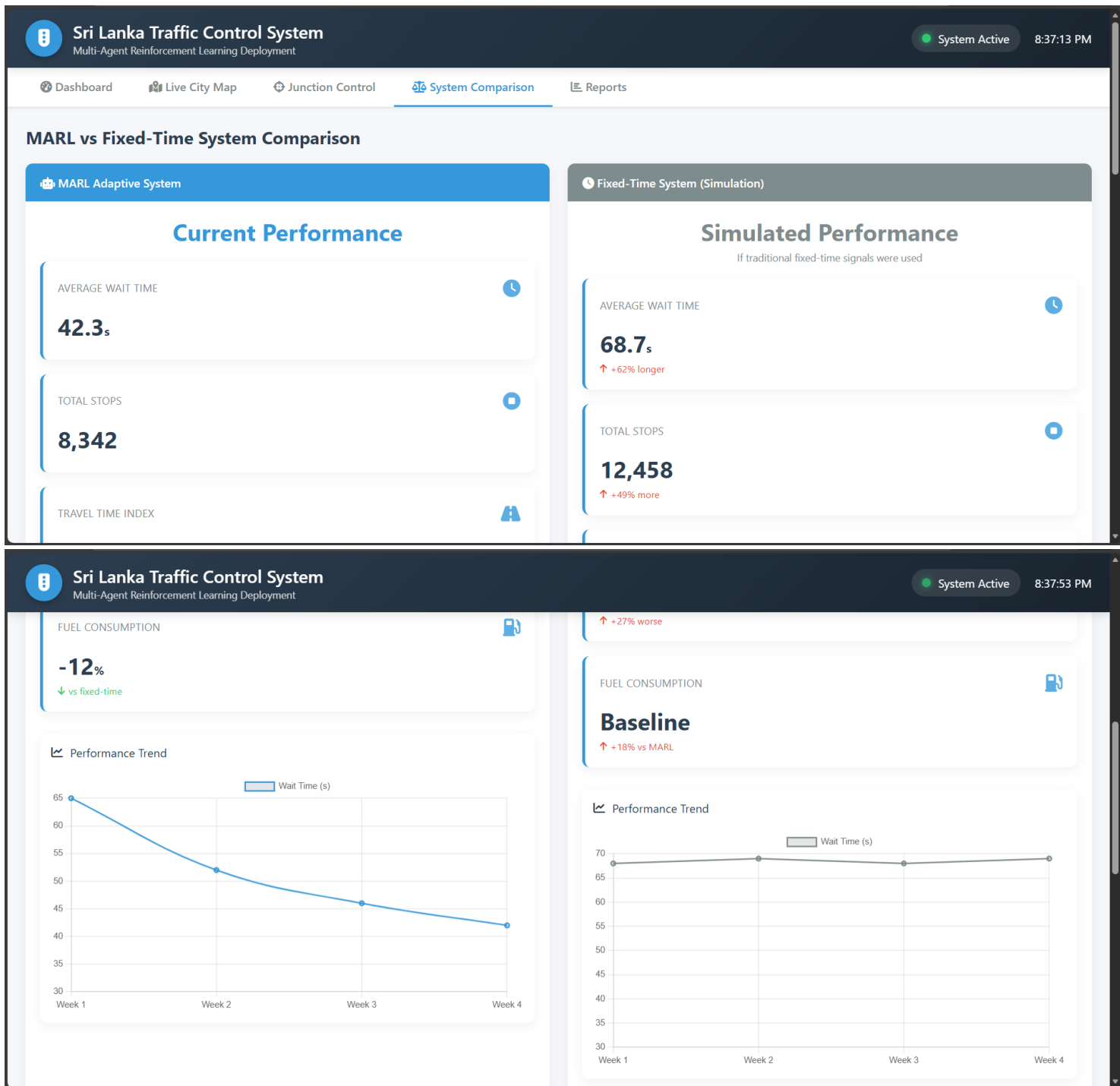


4. Initial Figma Design

a. Dashboard - to display overall city wide traffic analytics and statistics



b. System comparison - to compare our MARL traffic control and fixed timed traffic control for the operator to understand the ideal system to deploy



5. Feedback

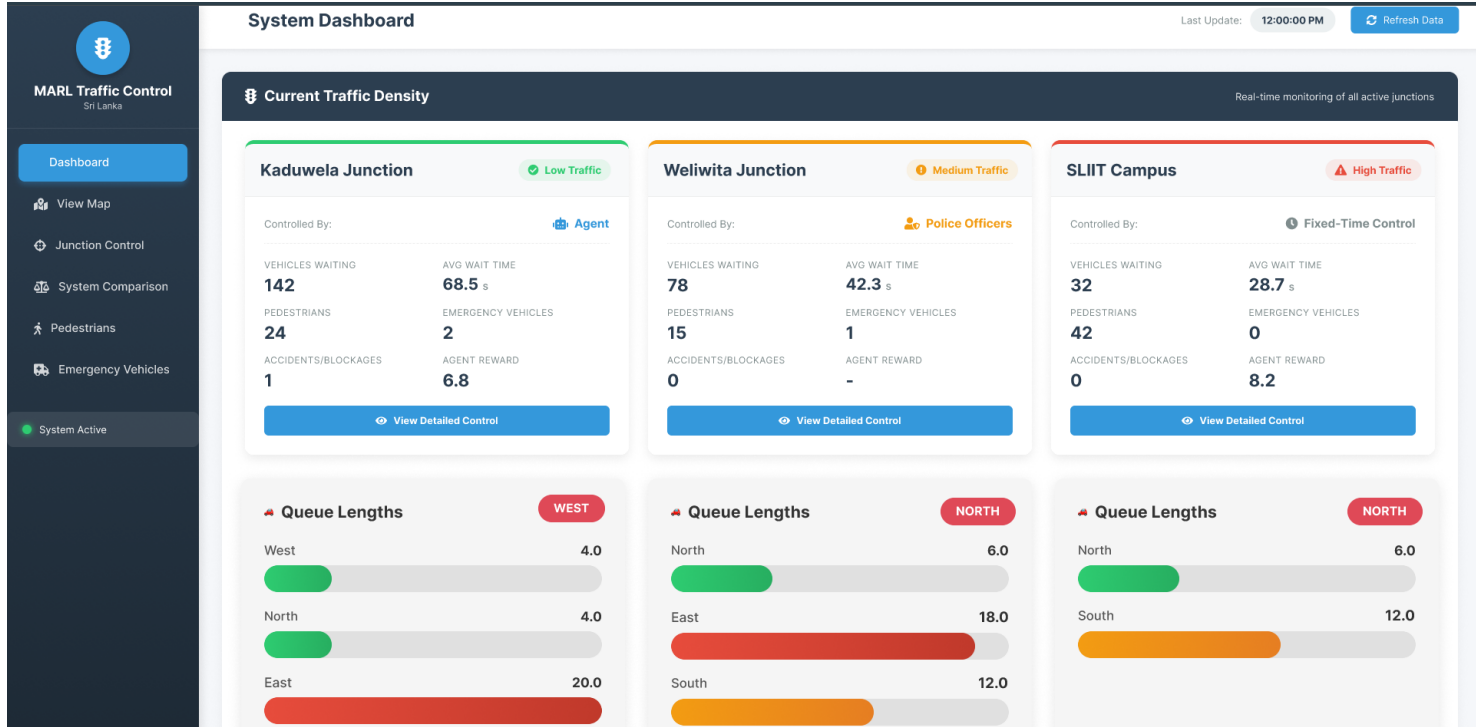
- The junction specific graphical analytics have been duplicated in a junction control page as well.
- Data oriented junction specific details on the dashboard are not user friendly. There should be a visual representation of the junction specific vehicle queue lengths.
- In system comparison, its lacking user friendliness and unable to grasp quick analytics and delay the purpose of the functionality which is to decide to which system the junction should switch under any circumstances
- System comparison may have another system called MARL without inter-communication to gain more options to switch into since it (MARL without inter-communication) may

- e. reduce traffic blockages in some scenarios.

6. Improvements - Final Figma UI

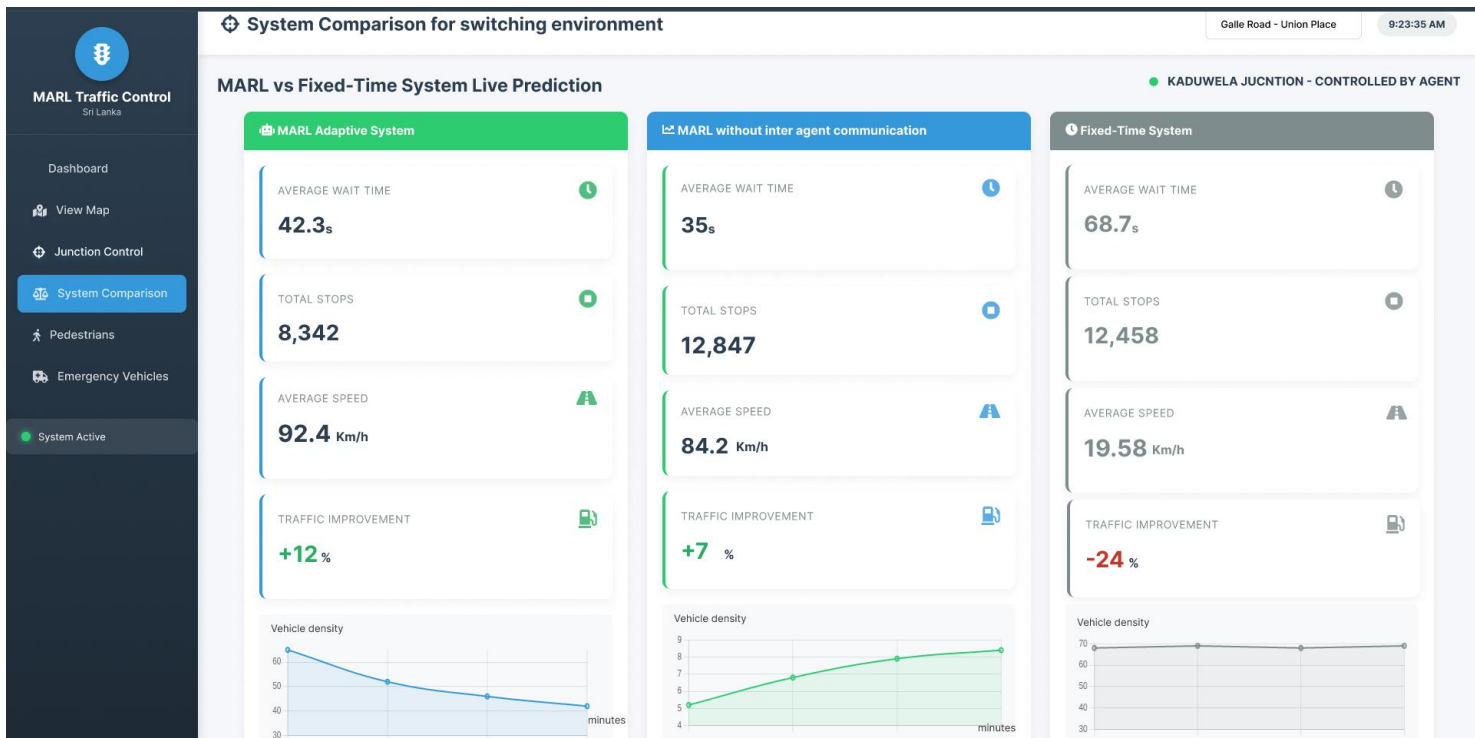
a. Dashboard - Improvements done:

- i. Removed the graphical statistics like charts in the dashboard for each junction specific cards since it is being duplicated back again in junction control page
- ii. Added visually apprehendable junction specific data which is queue length of vehicles per each lane at the intersection.



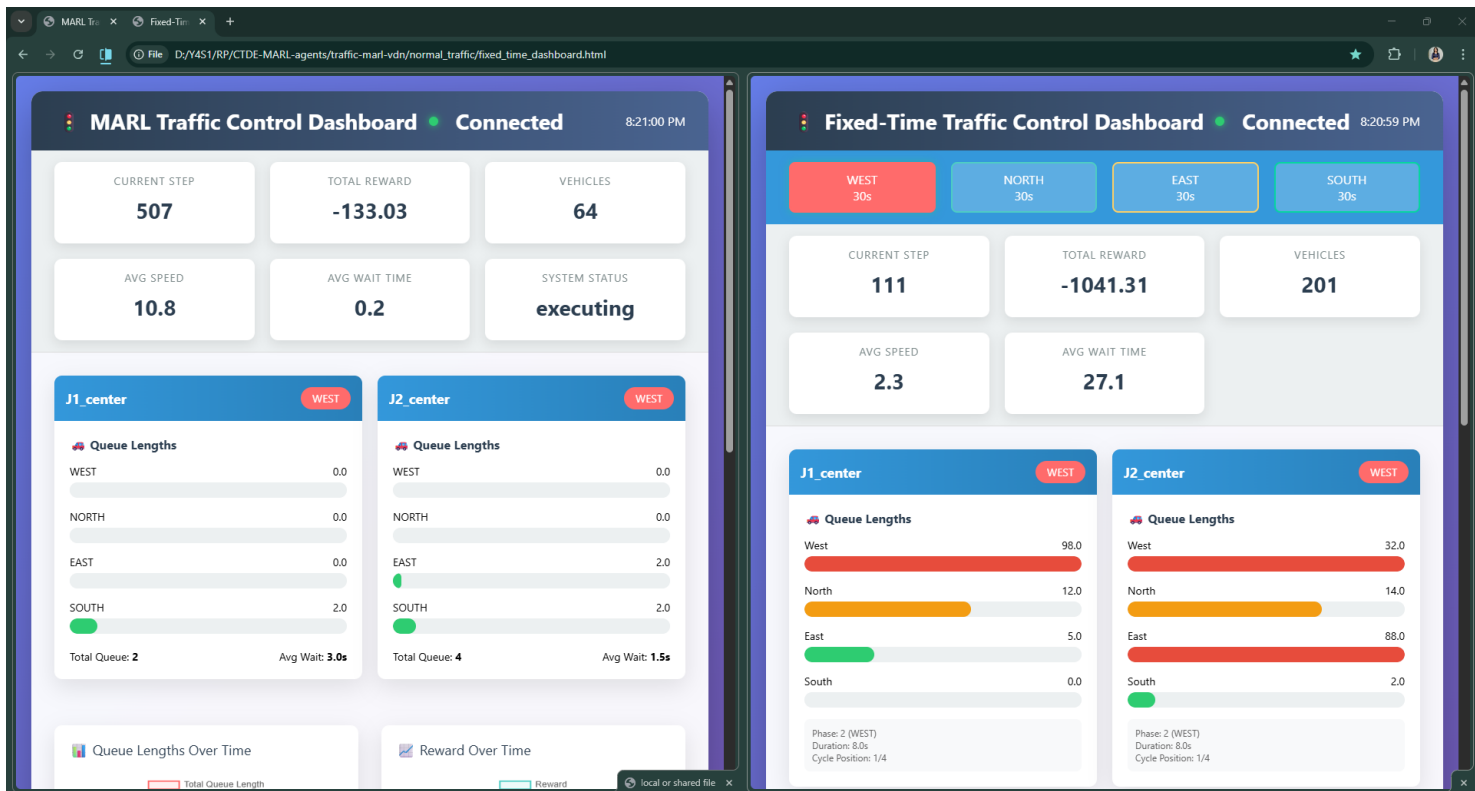
b. System Comparison - Improvements Done:

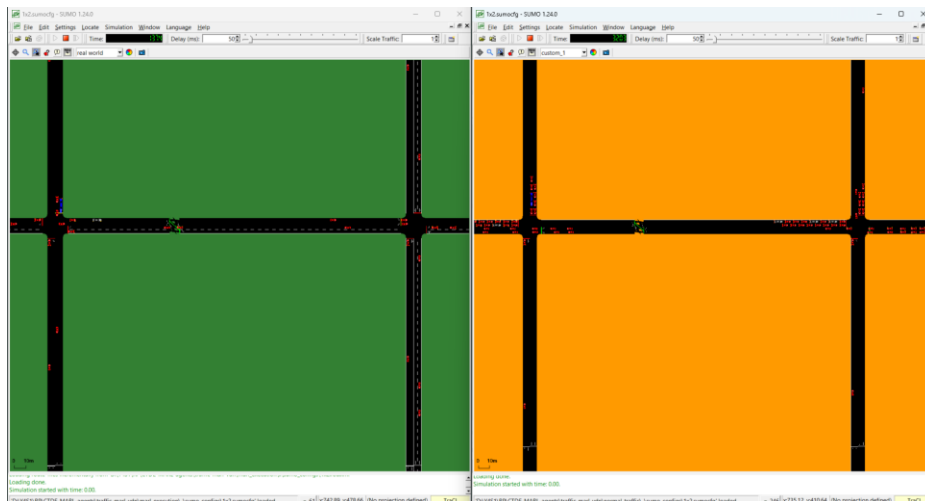
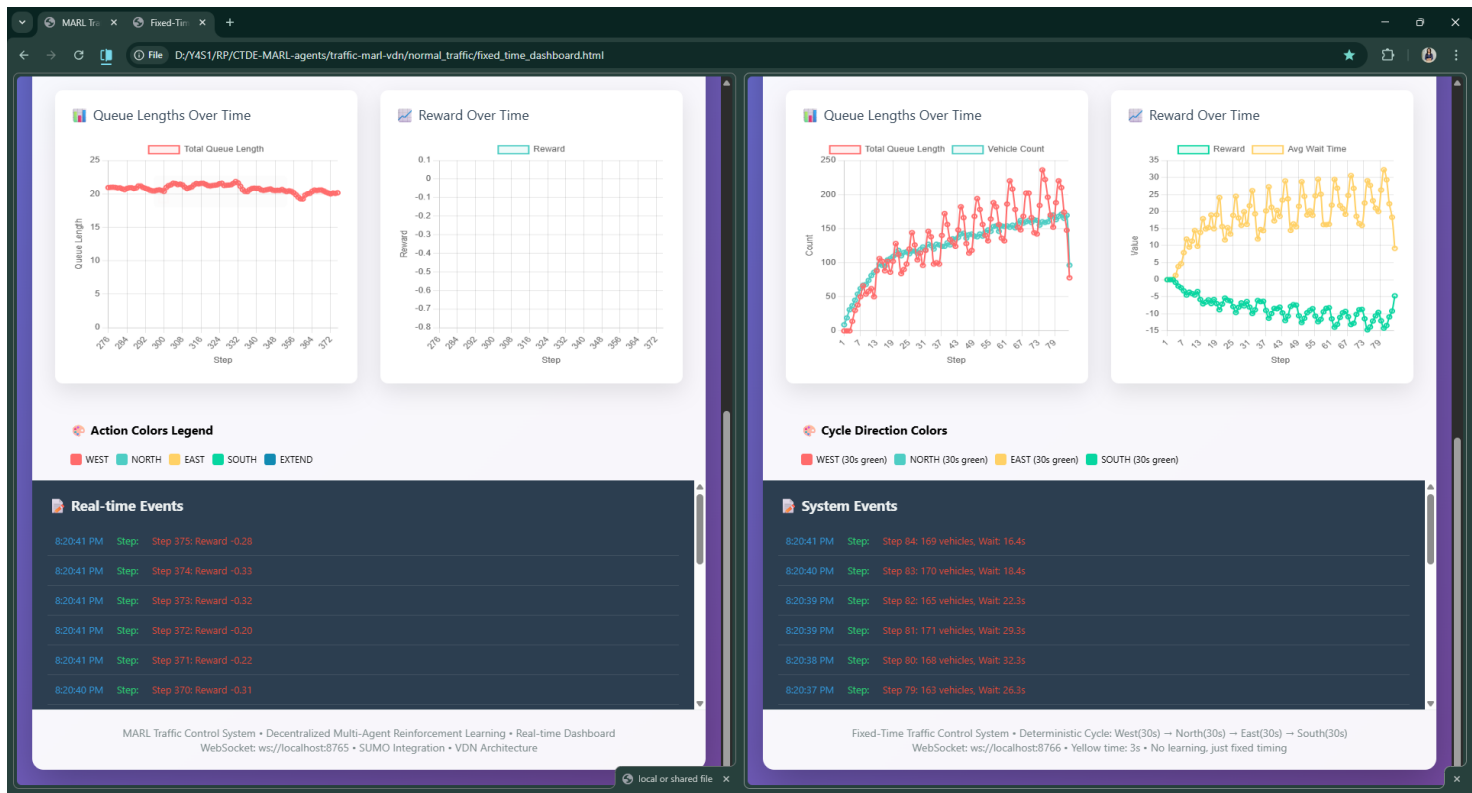
- i. Added user friendly interface by colors and icons for better grasp of information before switching environments
- ii. Added a 3rd environment called MARL without intercommunication, providing more options to switch into.



7. Dashboard for research progress - Final Real User Interface implemented

The MARL traffic control was compared against fixed timed traffic control at research progress presentation 1 held on 6th of january 2025. Real time data from SUMO was fetched via TraCi API through ZeroMQ to the relevant dashboards.





The following shows the two simulations done on SUMO for MARL traffic control vs fixed timed control. (Green background window - MARL traffic control, yellow background window - fixed timed traffic control). This is where the data was fetched from to the relevant dashboards above.